

EART 26001
Energy and Resources

Dr Stefan Schroeder

Lecture 8-1: Petroleum

Energy and Resources
Learning goals week 8

This week you will learn to:

- **Describe the uses and chemical properties of petroleum (hydrocarbons).**
- **Describe the elements of the petroleum system, and the geological constraints for the various elements.**
- **Recognize the exploration and production methods for petroleum recovery.**

Petroleum applications

- Fuel for domestic and industrial energy consumption
- Plastics
- Raw material for pharmaceutical products









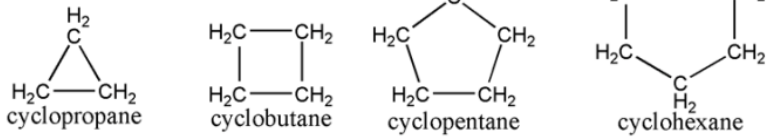




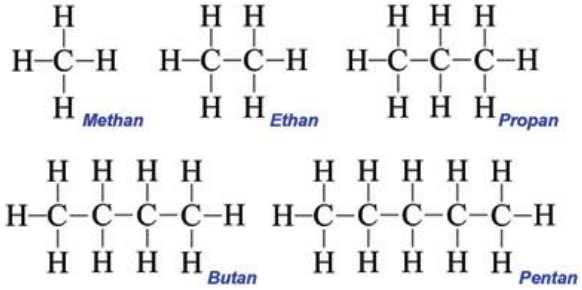
Properties of petroleum
Organic geochemistry

- Mix of different organic compounds
  - Largely simple and complex hydrocarbons
  - Other organic compounds containing S, O, N, some metals
- Majority fall into small range:
  - 30-60% Cycloalkanes ("Napthenes"): one or more C rings to which H atoms are attached (C_nH_{2n})
  - 15-60% Alkanes ("Paraffins"): single C-C bond chains to which H atoms are attached C_nH_{2n+2})
  - 3-30% Arenes ("Aromatic Hydrocarbons"): benzene rings (C_6H_6) to which other hydrocarbon groups may be attached
  - Anything else are usually Asphaltenes – complex organic compounds containing C, H, N, O, S, V, Ni (C:H ratio around 1:1.2)

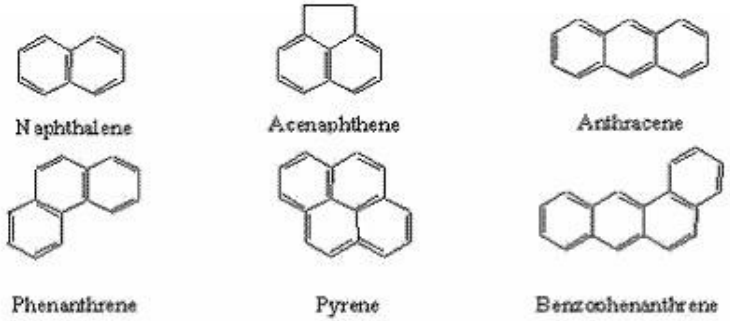
First four cycloalkanes



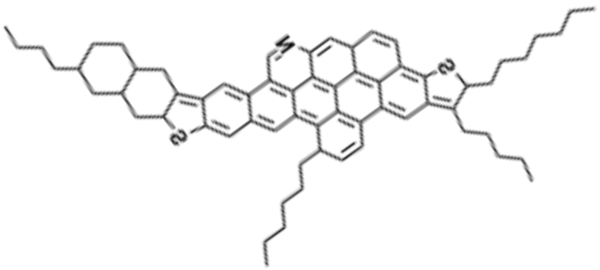
First five alkanes



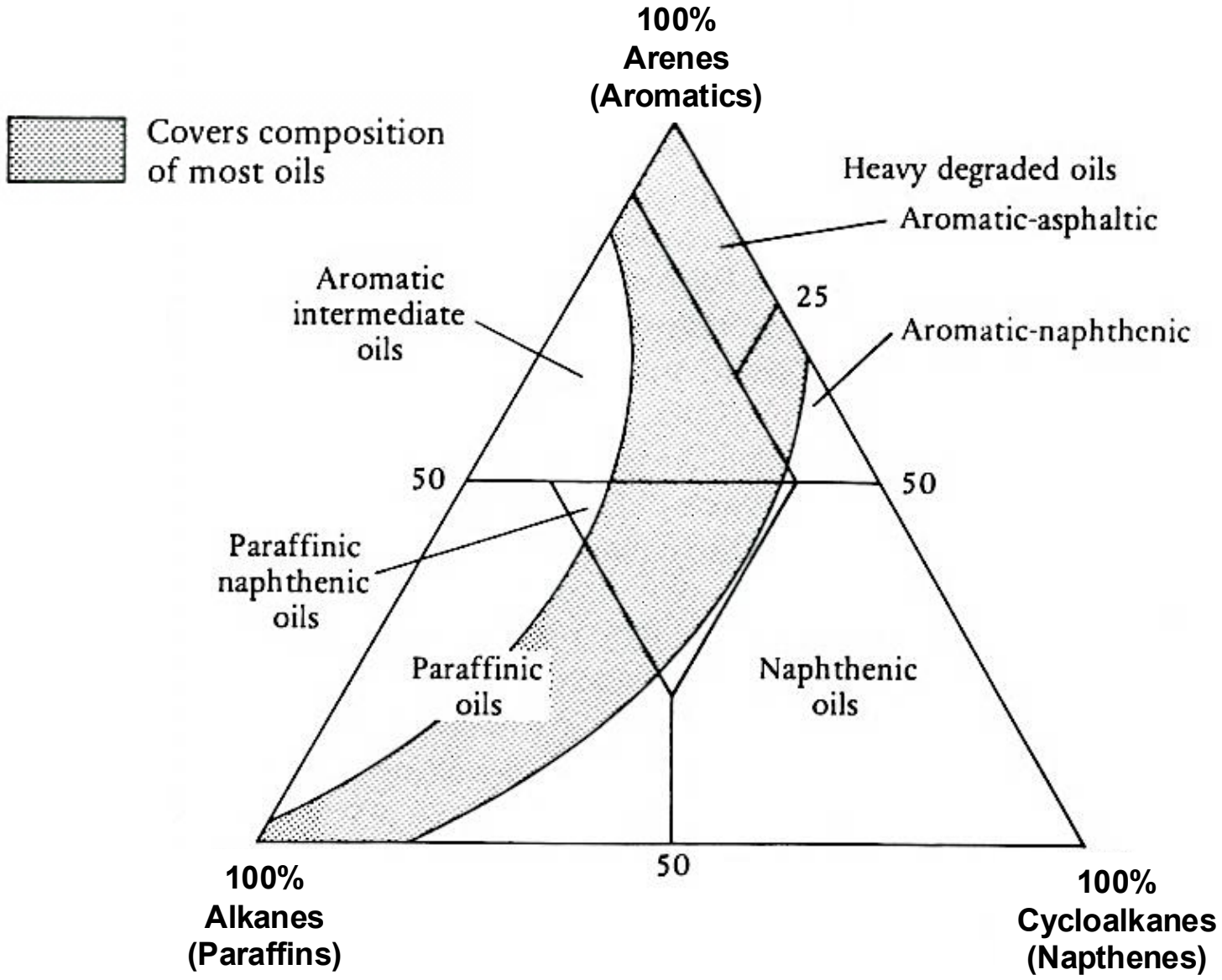
Examples of arenes



Example of an asphaltene



Properties of petroleum
Organic geochemistry



Properties of petroleum
Organic geochemistry

- Variable petroleum phases at surface and subsurface, depending on boiling points of organic compounds
- Crude oils are classified according to the American Institute of Petroleum (API) Gravity unit

$$API\ Gravity_{Petroleum} = \frac{141.5}{Specific\ Gravity_{Petroleum}} - 131.5$$

Degree API	Specific Gravity	Density [kg/m³]
8	1.014	1012
9	1.007	1005
10	1.000	998
15	0.966	964
20	0.934	932
25	0.904	902
30	0.876	874
40	0.825	823
50	0.780	778

← 10˚ API - water

← 22˚ API

← 30˚ API

↑ Heavy oils

Medium oils

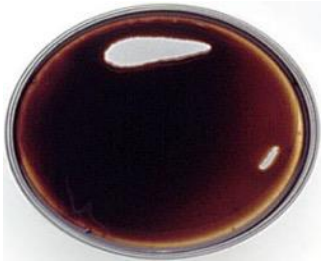
Light oils ↓



Gas



Light oil



Medium oil



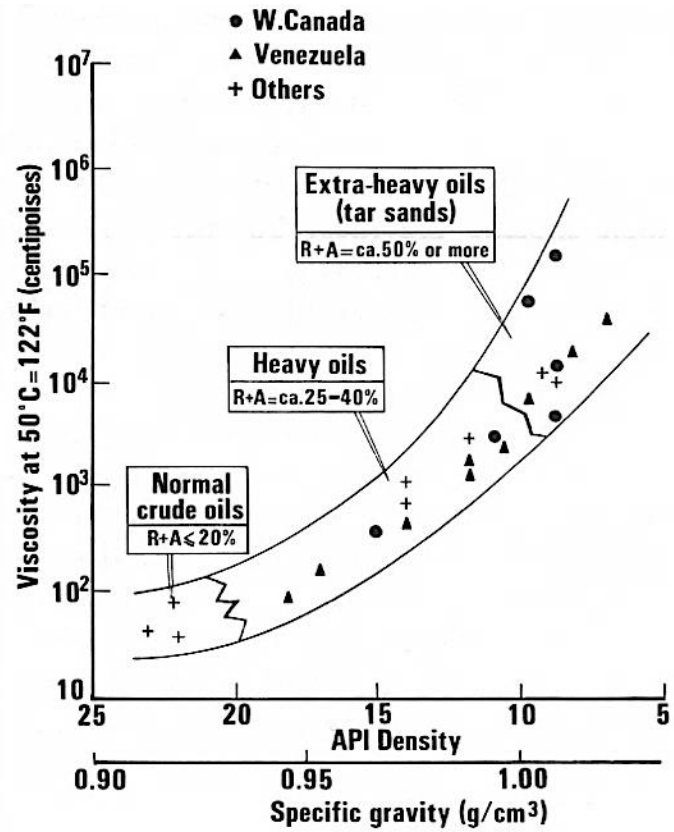
Heavy oil

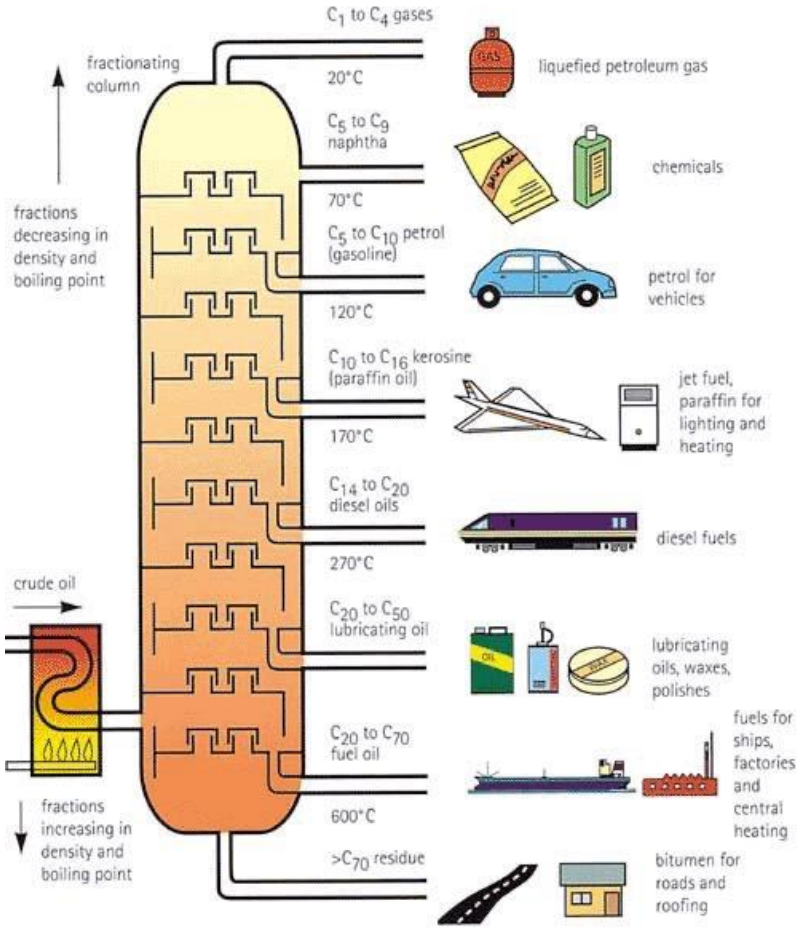


Bitumen

Properties of petroleum
Organic geochemistry

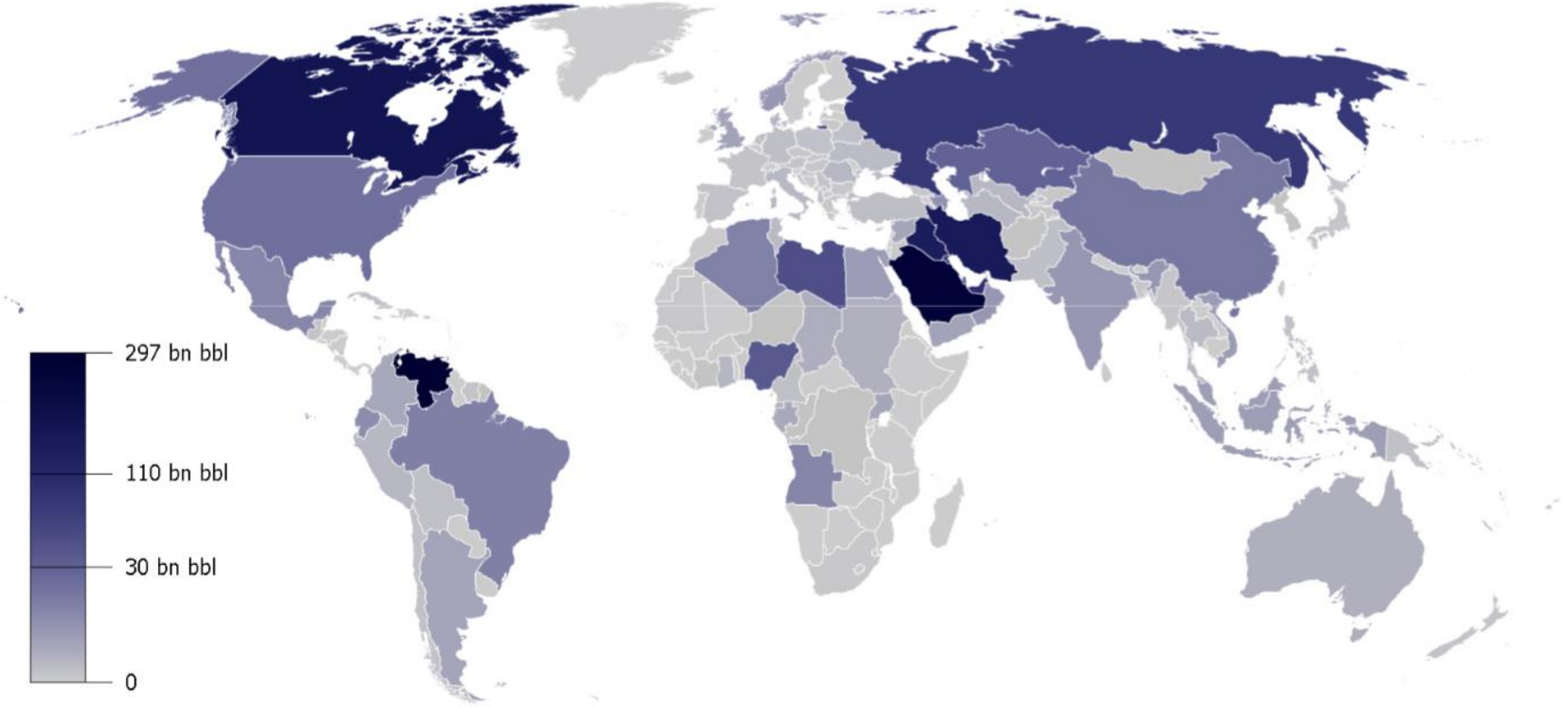
- As API density decreases (specific gravity increases), C chain length increases, viscosity increases
- Low API density crude oils are more viscous, harder to extract, require cracking for commercial use
- High API density crude oils (and gas) are less viscous, easier to extract, have more commercial uses with less cracking required





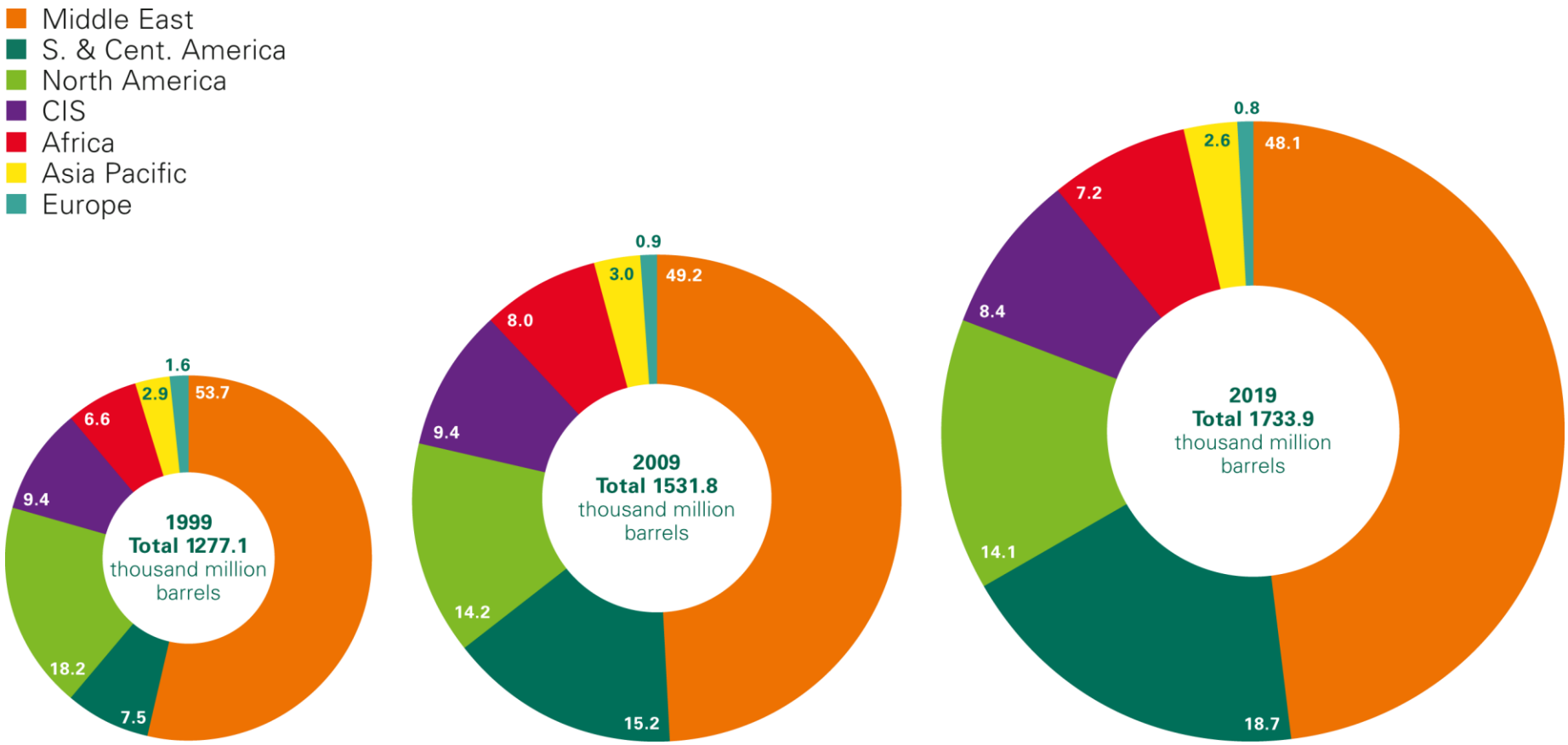
Petroleum
Reserves and demand

Proved oil reserves, 2013 (Source: World factbook)



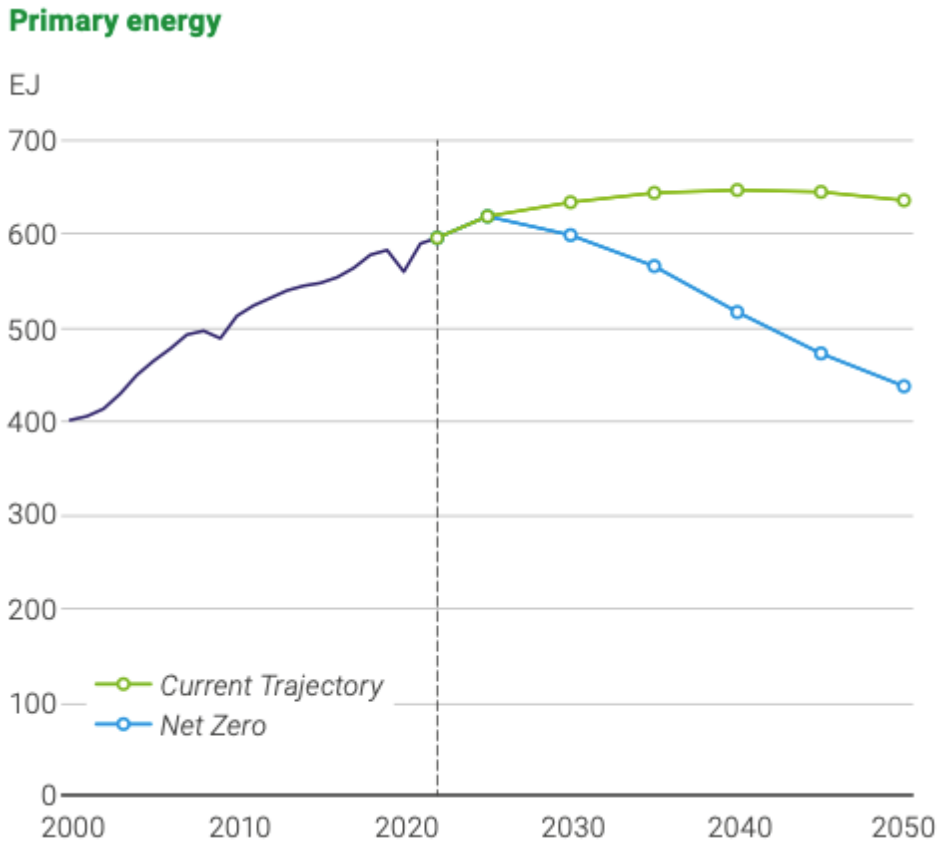
Petroleum
Reserves and demand

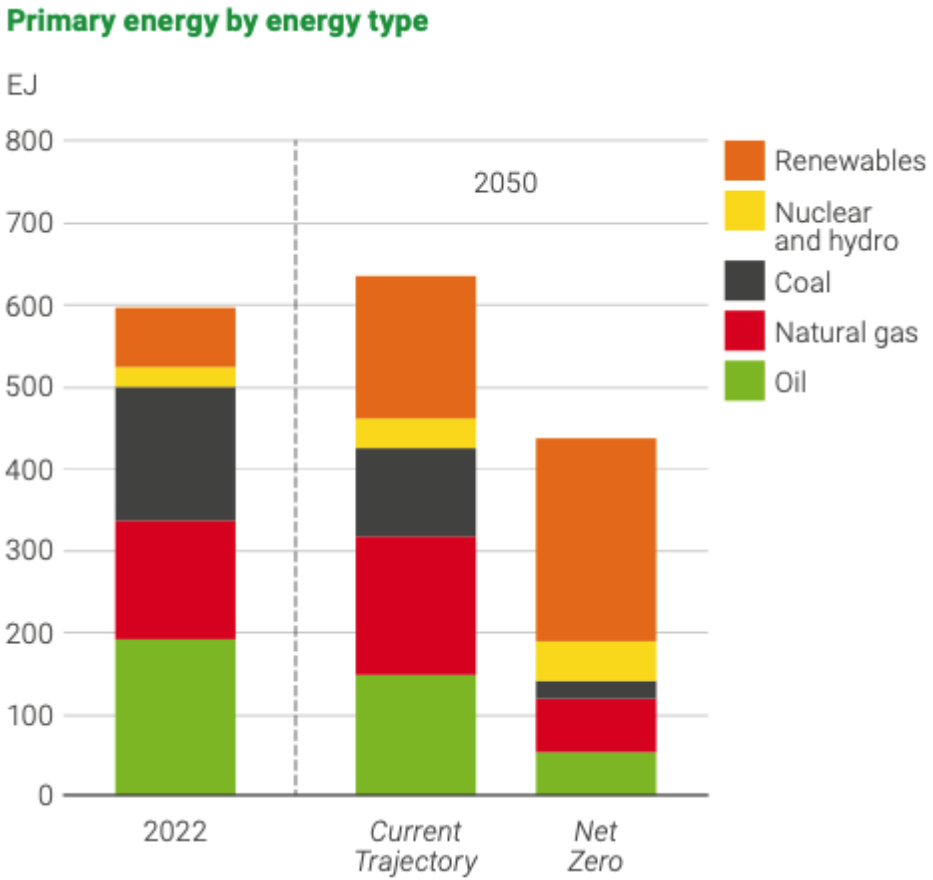
Proved oil reserves, 2019 (Source: BP)



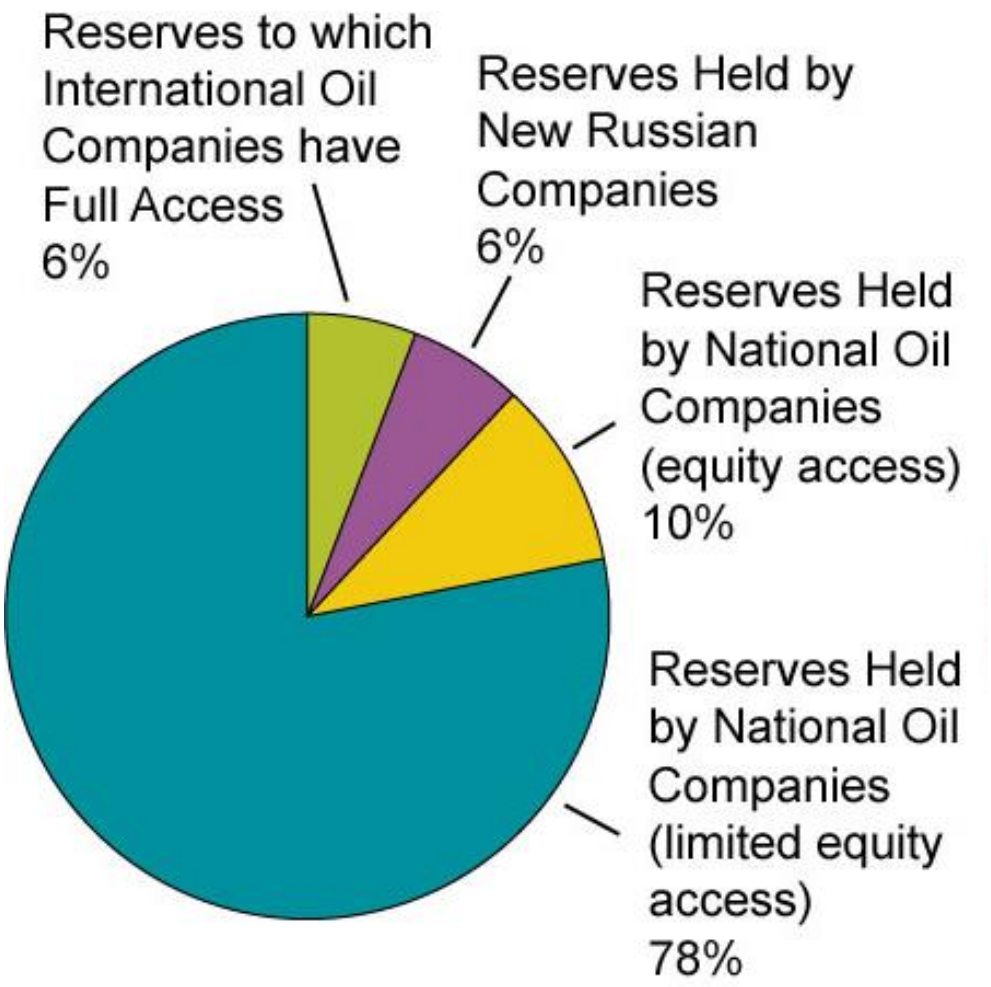
Petroleum
Reserves and demand

Projected energy demand (Source: BP Energy Outlook 2024)





Petroleum
Reserves



Source: PFC Energy, 2008

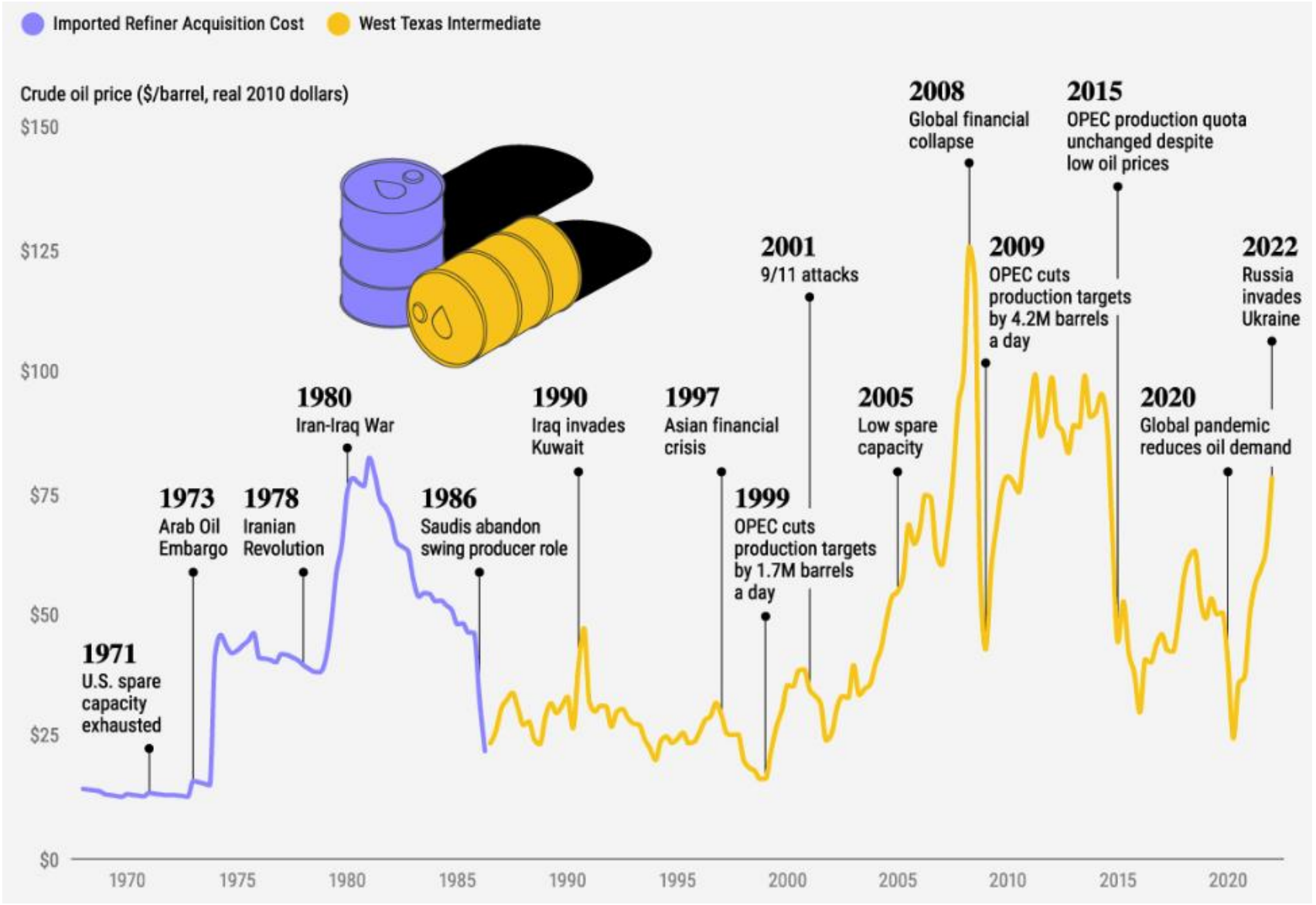
- **88% owned by nation states**
- **"Free market" companies have access to 16%**
- **6% held by new Russian companies (not really free market – "quasi-free")**
- **Even the world's largest public oil company, Exxon, has access to <1% of the World's oil**

<u>International oil company ('majors'):</u>
Owned by shareholders and operate internationally, often in partnership with NOCs;
eg Exxon, Shell, BP, Chevron, Total

<u>National oil company:</u>
Owned by the government
eg Saudi Aramco, Sonangol, PDO, PDVSA

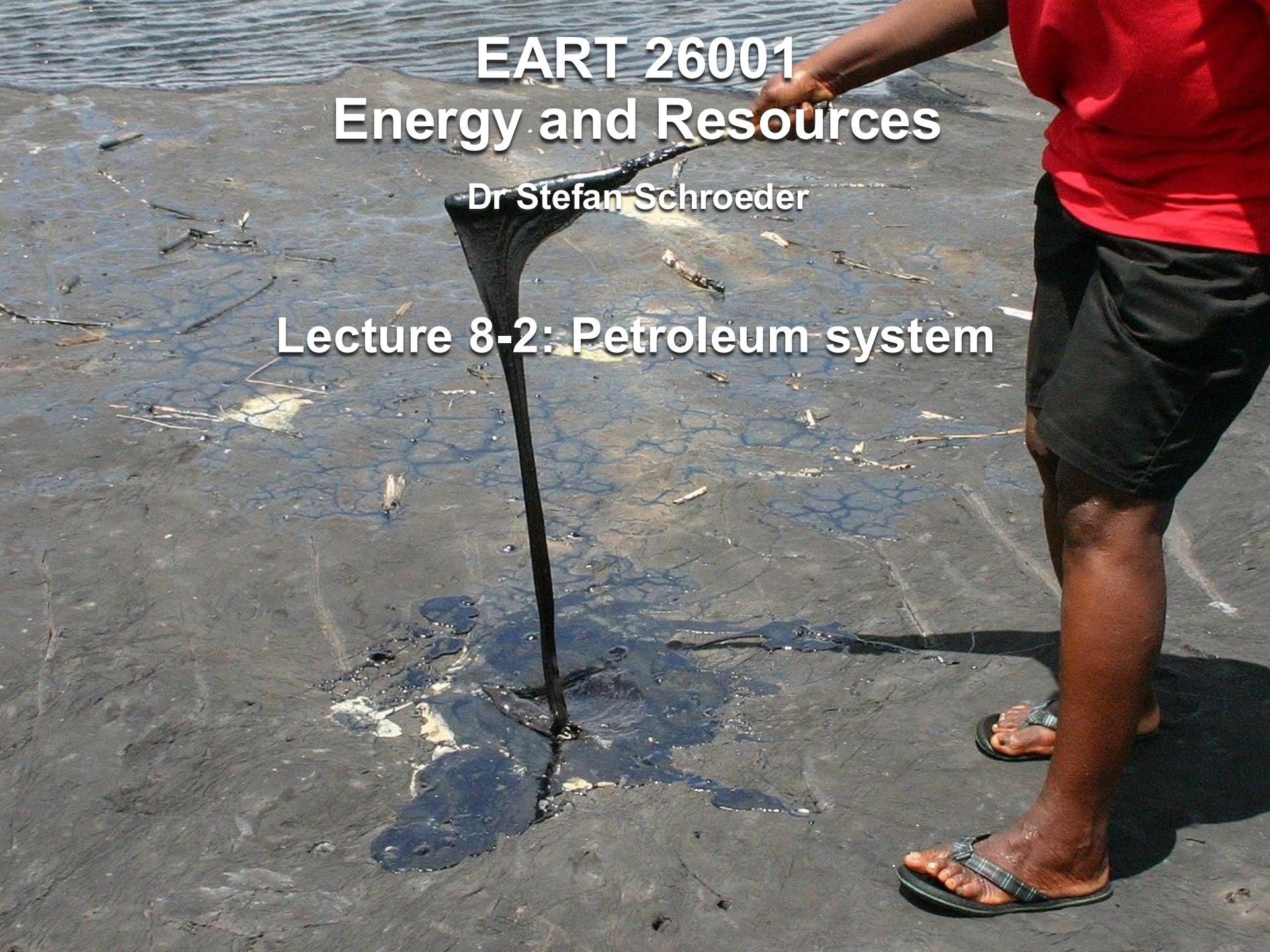
Petroleum
Market

Crude oil prices



Source: US Energy Information Administration

© www.visualcapitalist.com

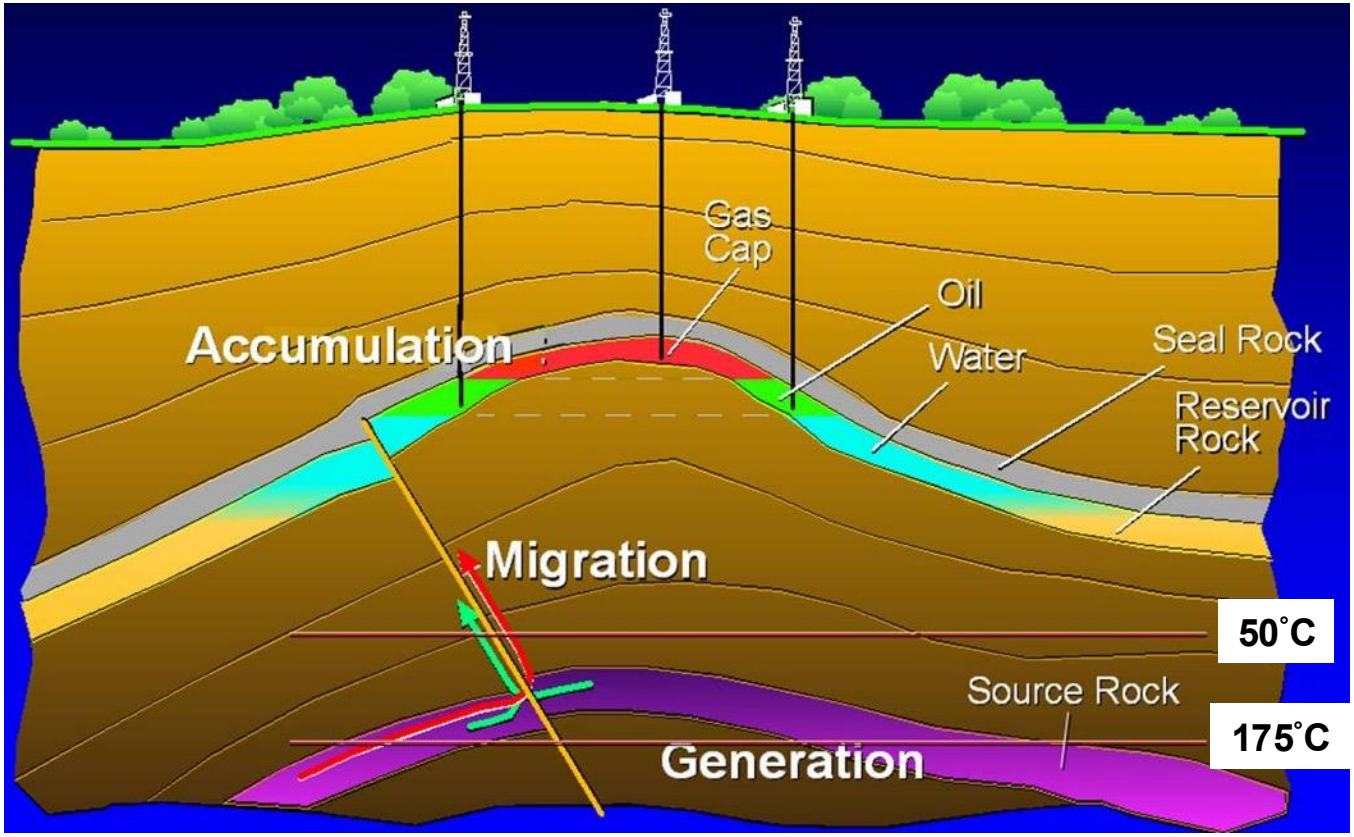


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Lecture 8-2: Petroleum system

Petroleum system



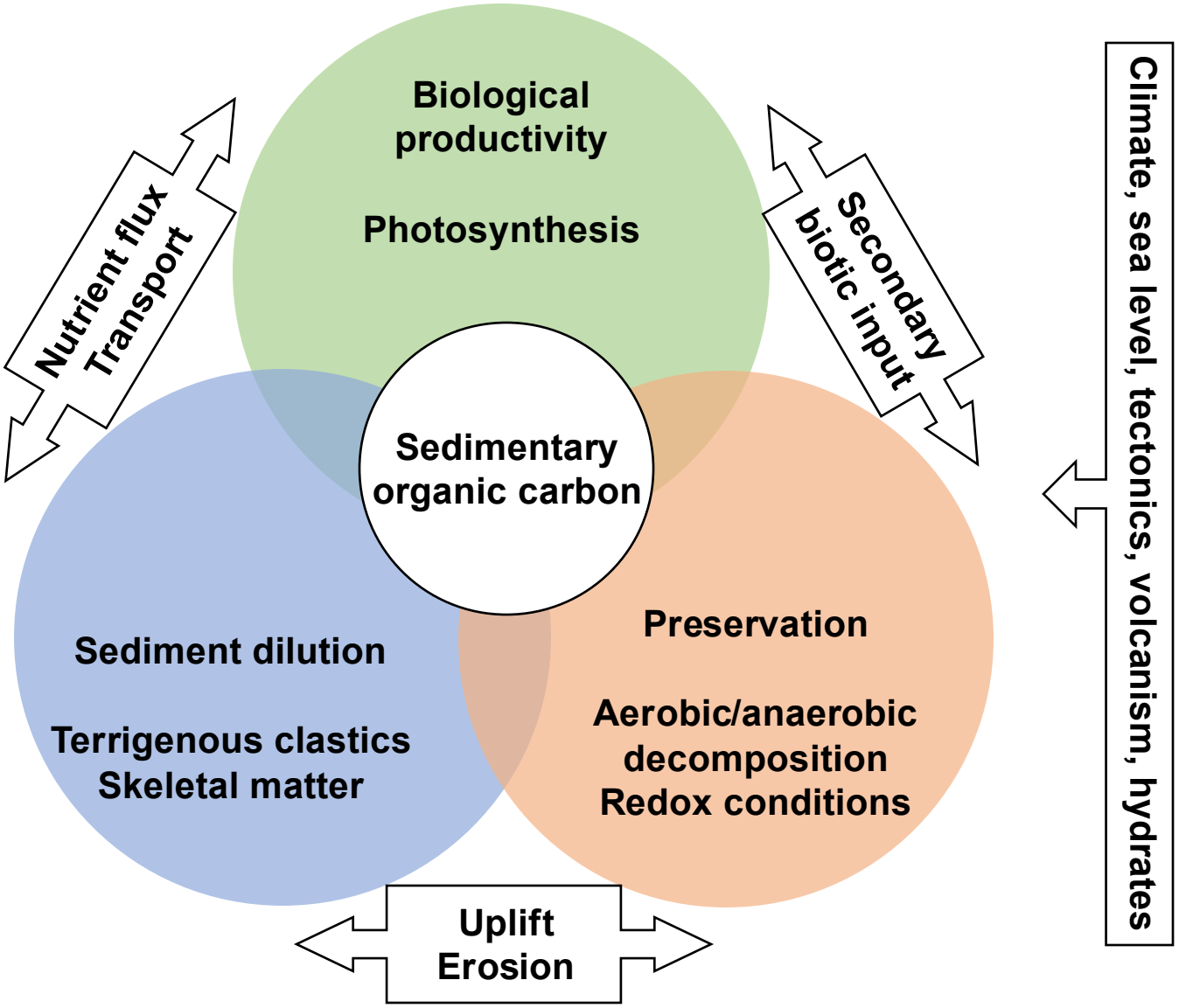
Necessary elements:

- Source rock
- Reservoir rock
- Seal rock
- Overburden

Necessary processes:

- Trap formation
- Generation of hydrocarbons (by maturation)
- Hydrocarbon migration and accumulation in trap

Source rock
Accumulation of organic matter



Walters 2017

Source rock
Formation of organic matter



- Rock with >1% TOC (total organic carbon) from:
  - Proteins – animal tissue, enzymes
  - Carbohydrates – animal tissue
  - Lipids – algae, pollen, spores
  - Lignin – plant tissue

- Accumulation of organic matter to form source rock is controlled by rates of organic productivity vs organic matter degradation:

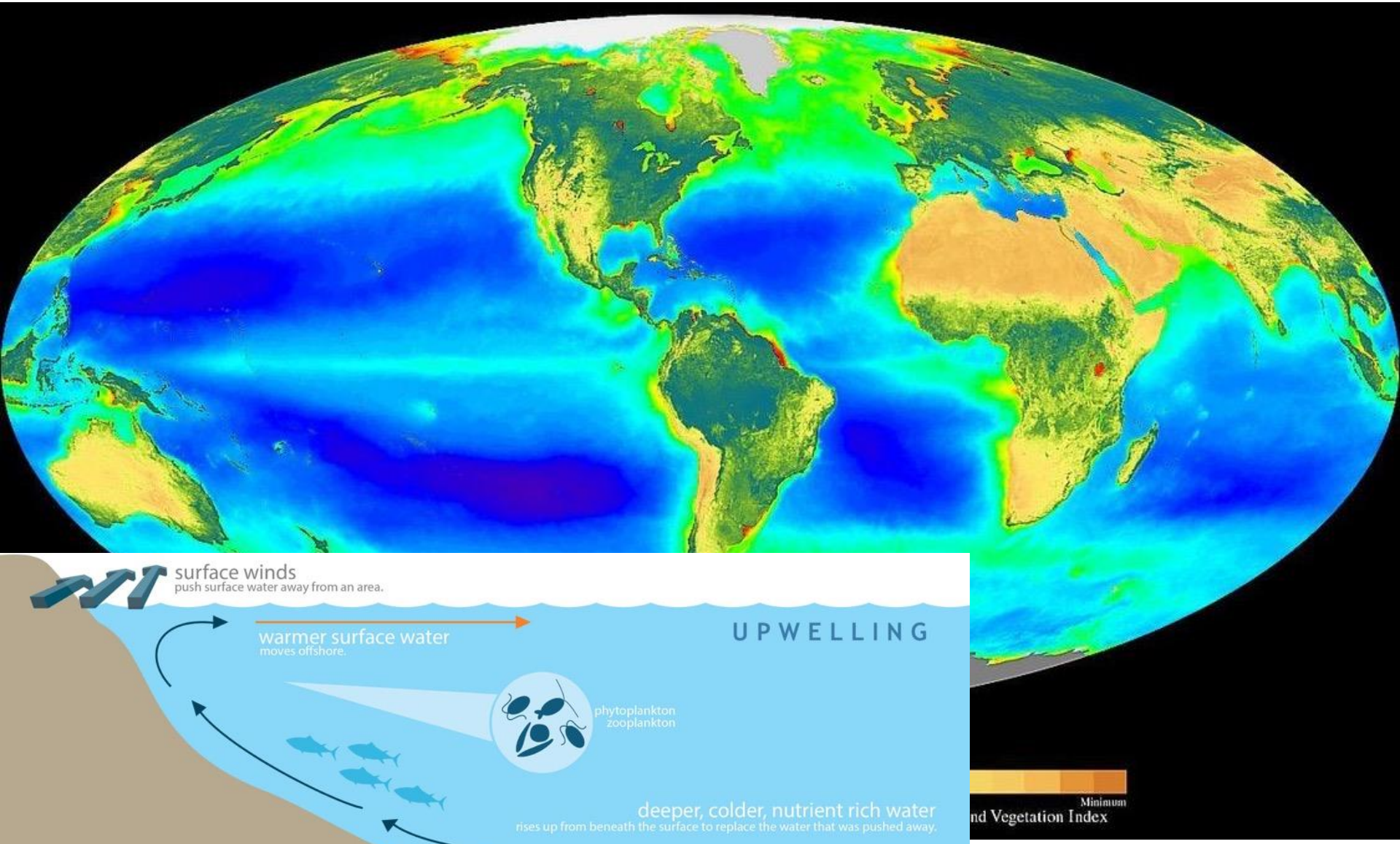
$$6CO_2 + 6H_20 \leftrightarrow C_6H_{12}O_6 + 6O_2$$

Carbon dioxide *Water* *Carbohydrate* *Oxygen*

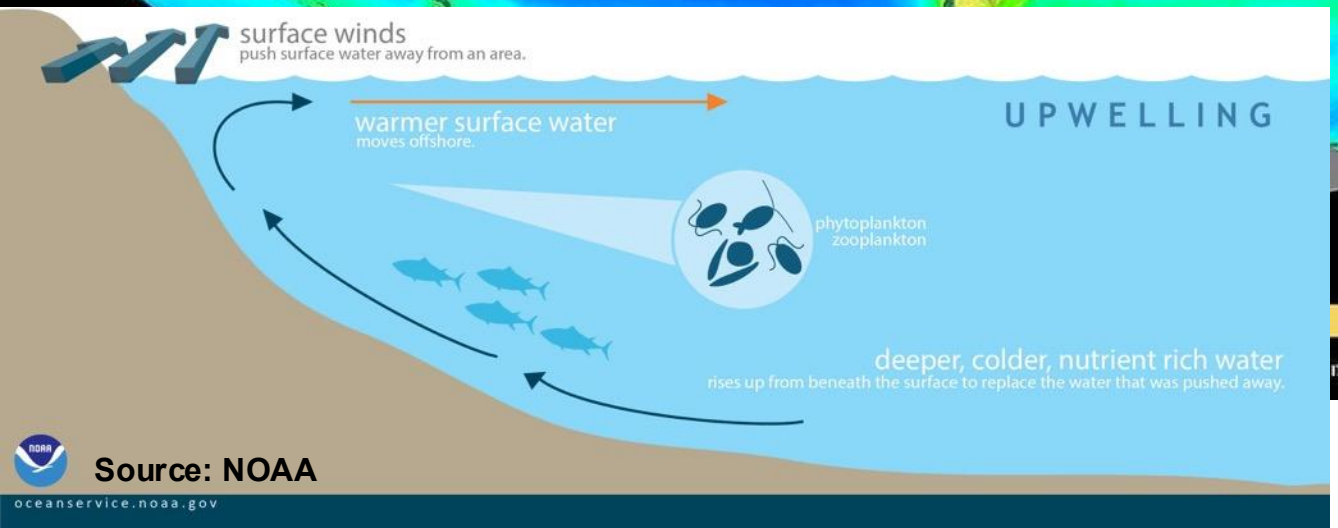
C fixation: productivity

C oxidation: respiration, re-mineralization, combustion

Source rock
Primary productivity

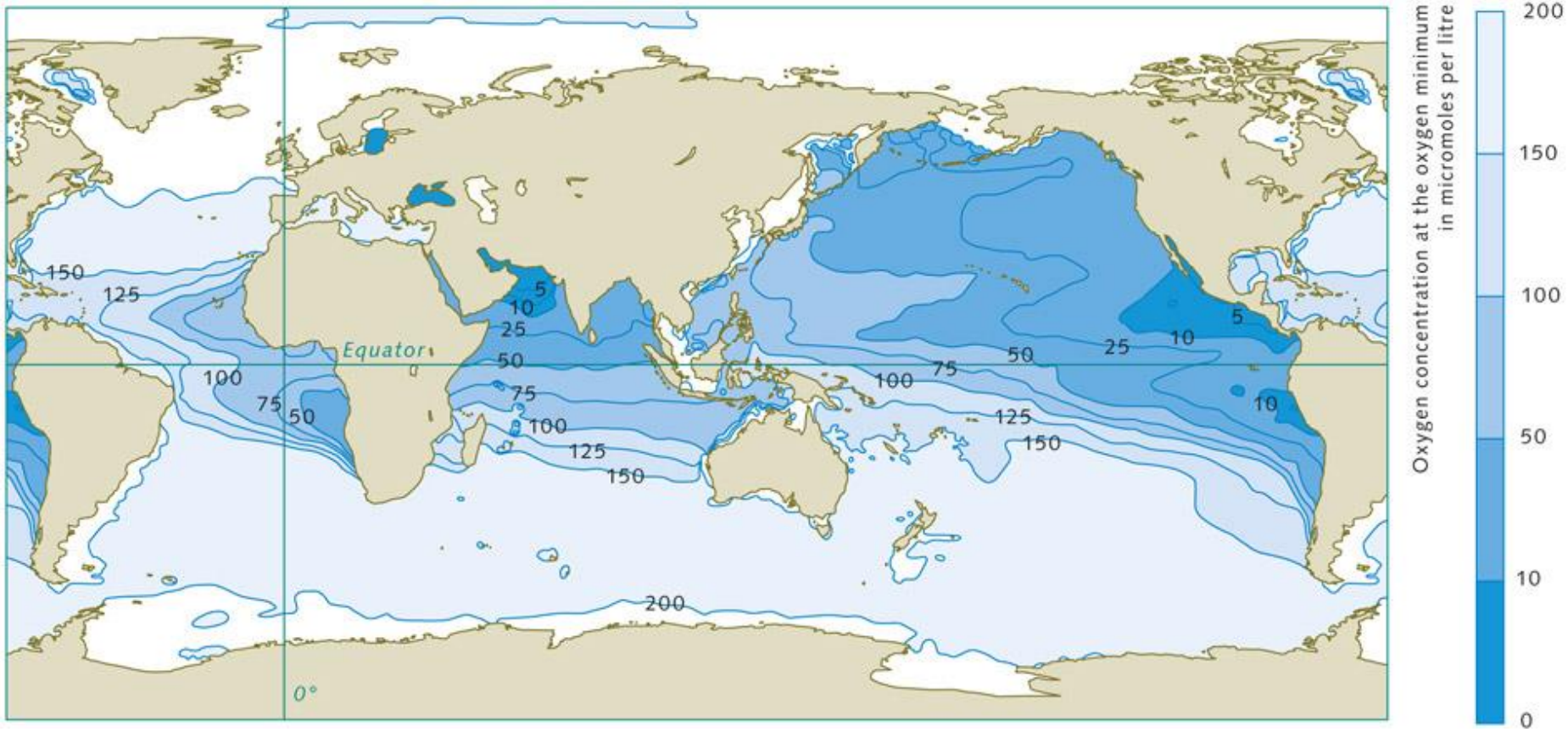


Source: NASA



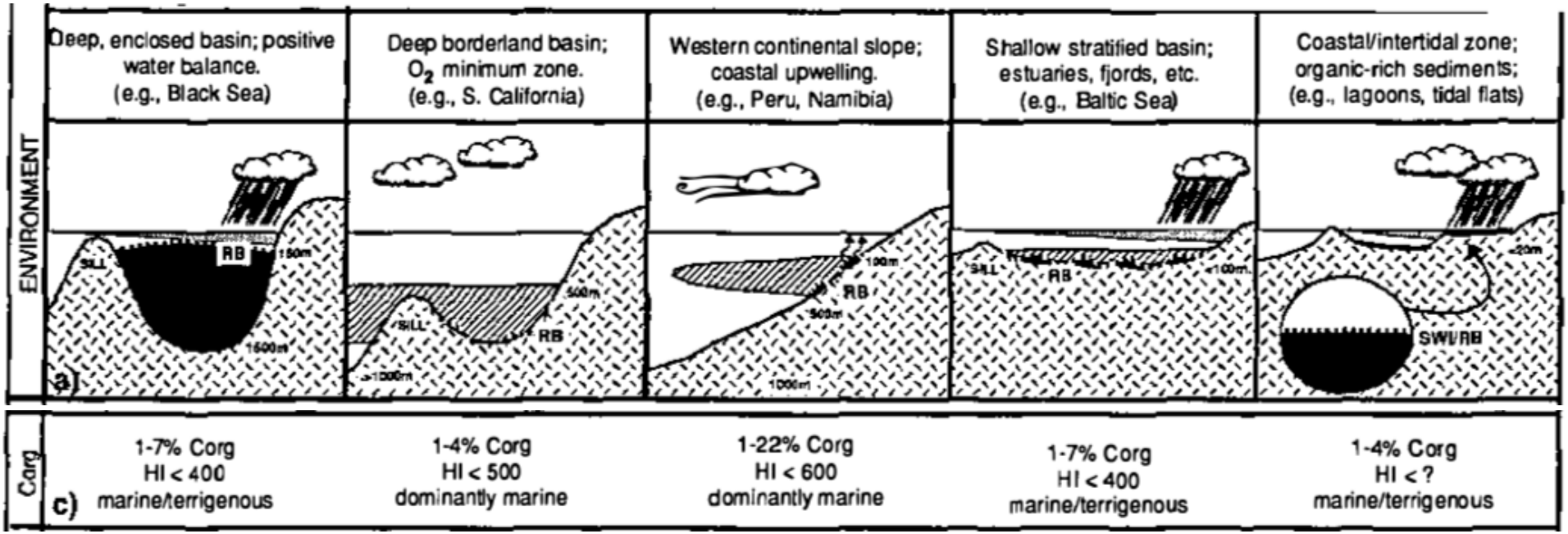
Source: NOAA

Source rock
Organic matter preservation



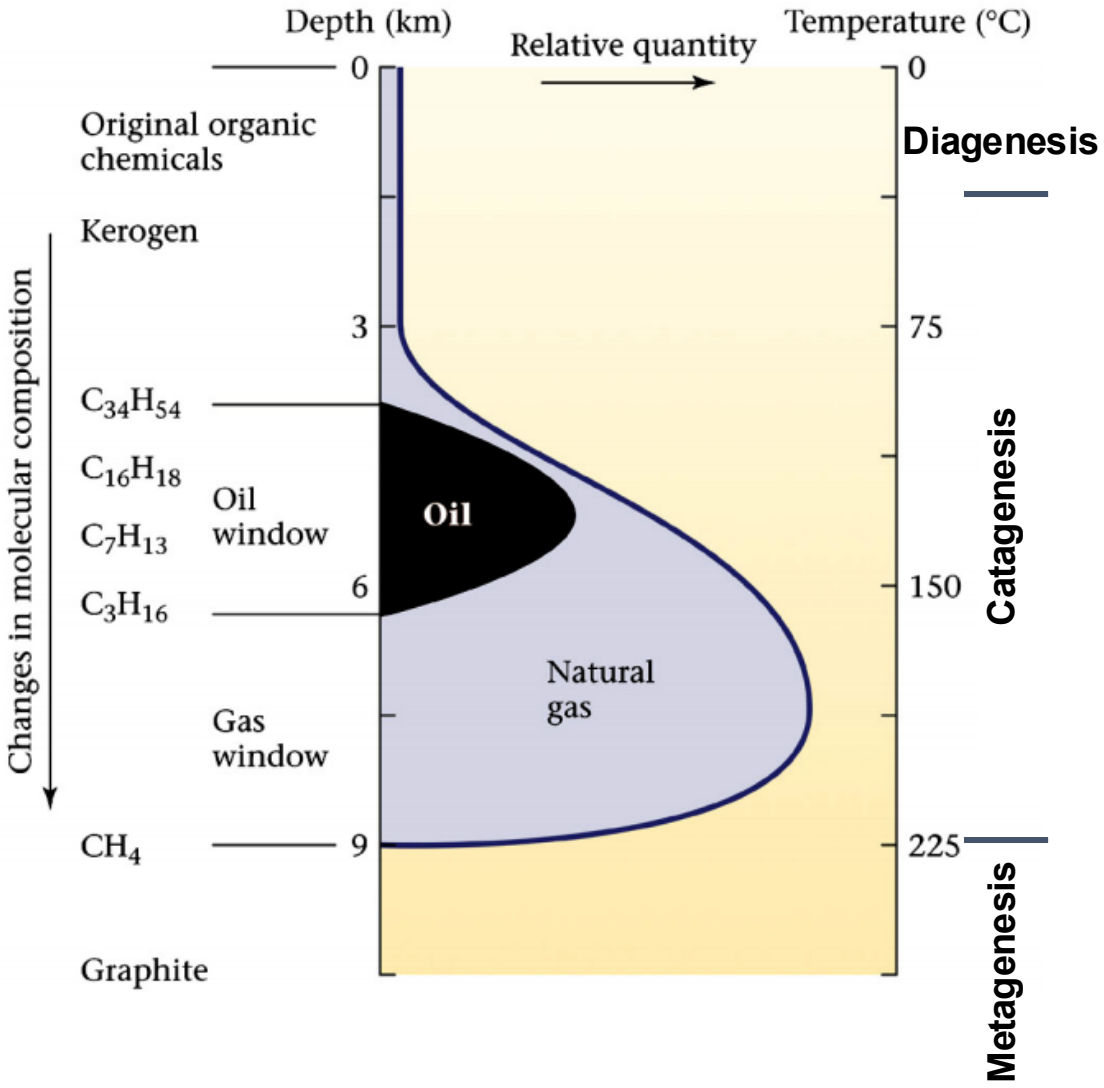
Source: NOAA

Source rock
Organic matter preservation



Arthur & Sagemann 1994

Source rock
Organic matter maturation



- Burial under increased overburden provides increase in temperature -> transformations of organic matter:

< 40-80°C: anaerobic and abiotic breakdown of organic matter – loss H_2O, CH_4, CO_2, O: kerogen

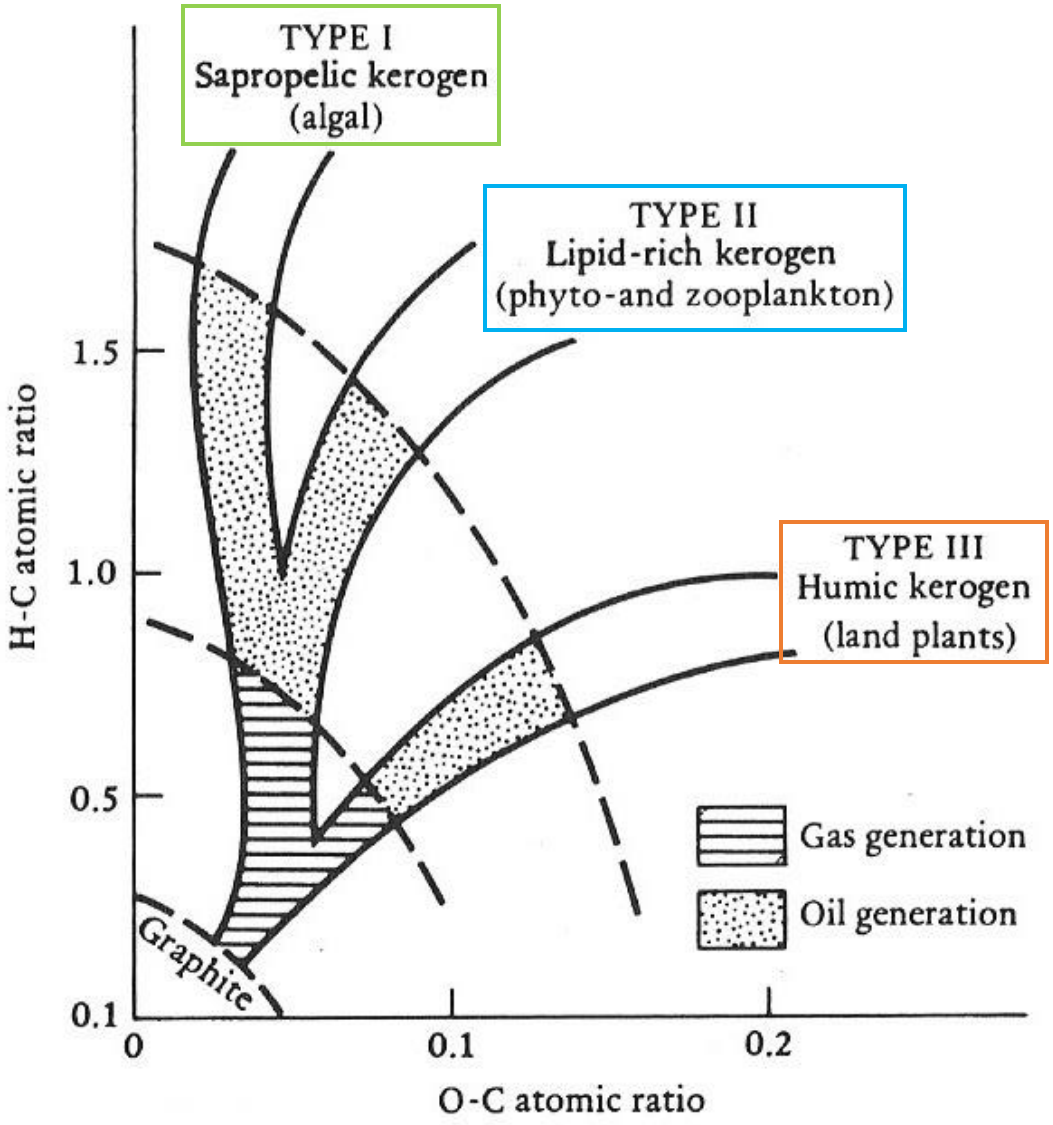
> 40-80°C: breakdown of kerogen

100-150°C: oil window - breakdown of C bonds, formation of shorter chains

150-200°C: gas window – depletion in H, leaves CH_4

>200°C: only graphite remaining - overmature

Source rock
Organic matter maturation



Selley 1998

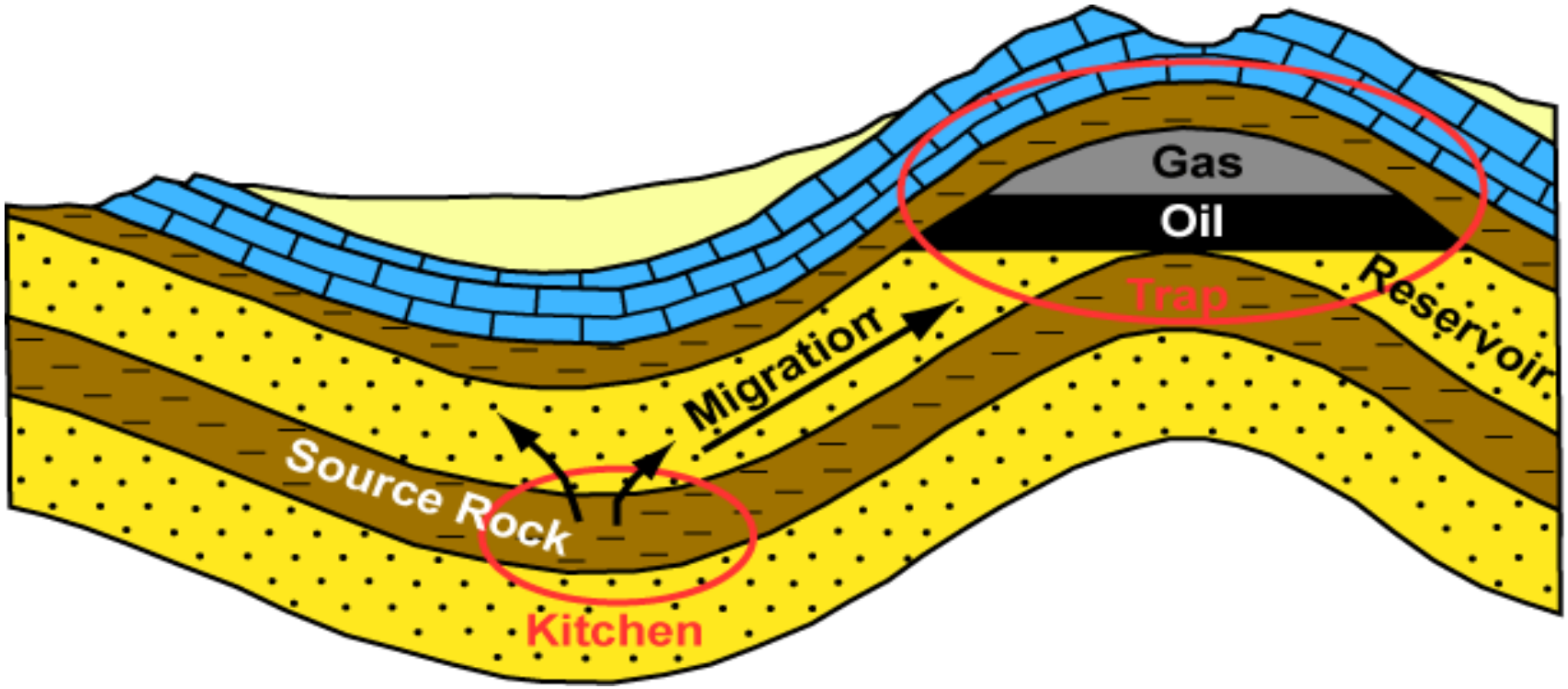
- **O, N, H_2O, CO_2 are released: convergence to high C and short C-H chains**

Type I kerogen:
- Algae
- Lipid-rich (fat, oil, wax)
- Mostly oil – oil shales (e.g. Eocene Green River Fm.)

Type II kerogen:
- Algae, zoo/phytoplankton
- Lipid-rich
- Oil and gas – marine source rocks (e.g. Jurassic Kimmeridge Clay)

Type III kerogen:
- Land plants
- Humic (lignin, tannin, cellulose)
- Gas – coals

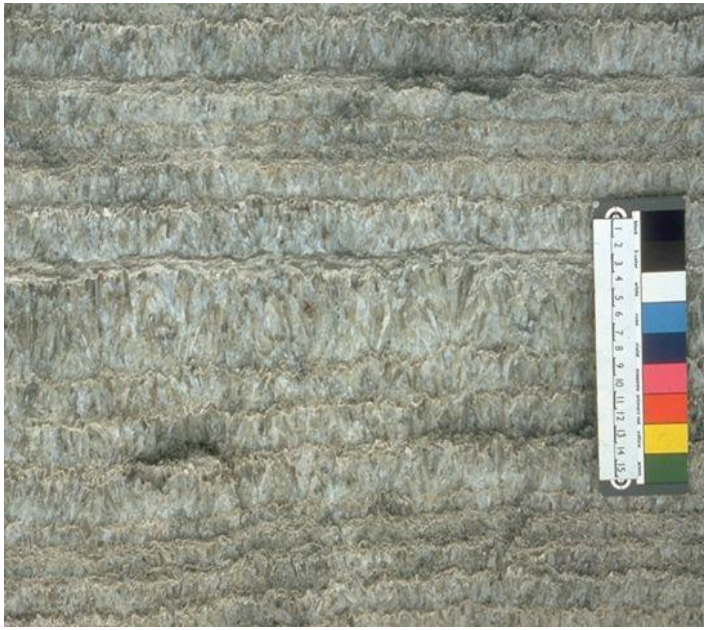
Accumulation



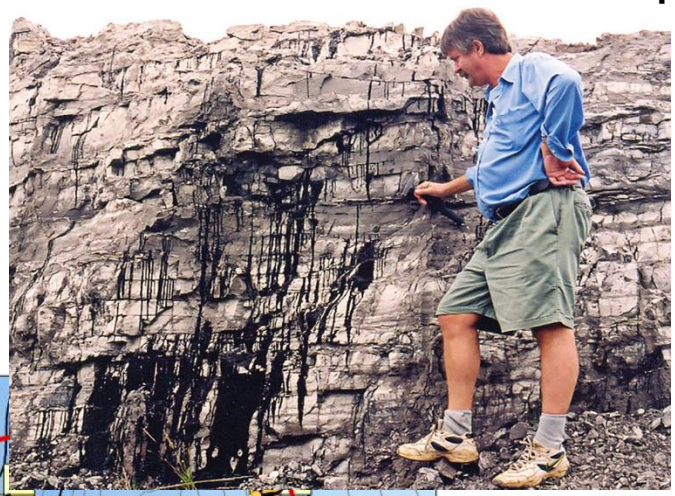
- Pressure drives expulsion of generated hydrocarbons, followed by migration due to buoyancy
- Migration depends on permeable carrier beds
- Accumulation requires trap with reservoir, below seal

Seal

- Any impermeable lithology above reservoir, that prevents further bouyant rise of hydrocarbons
- Seals may leak => surface seeps, detect accumulations

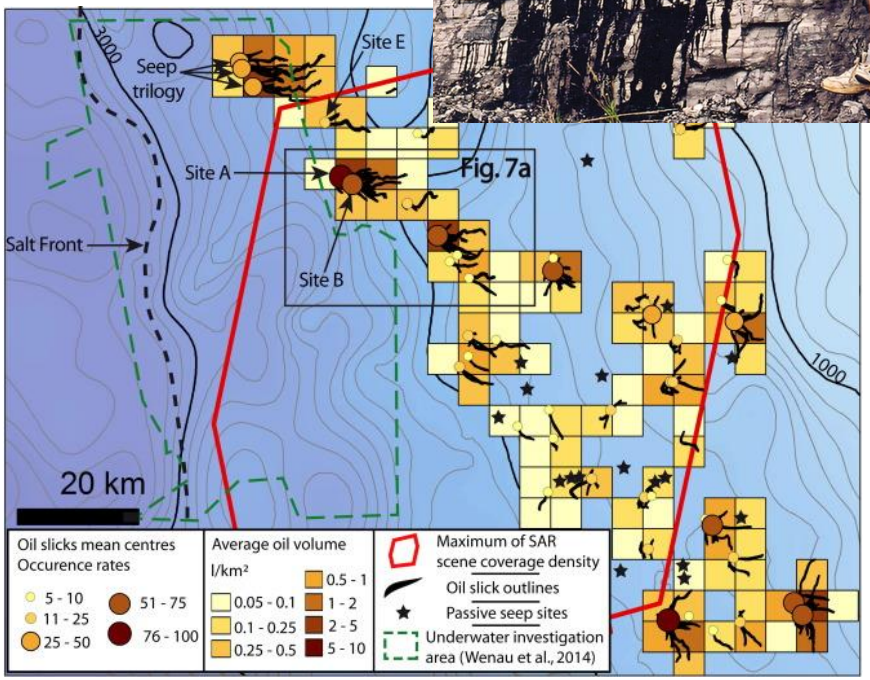


Surface seep



© Angola field group

Offshore seeps

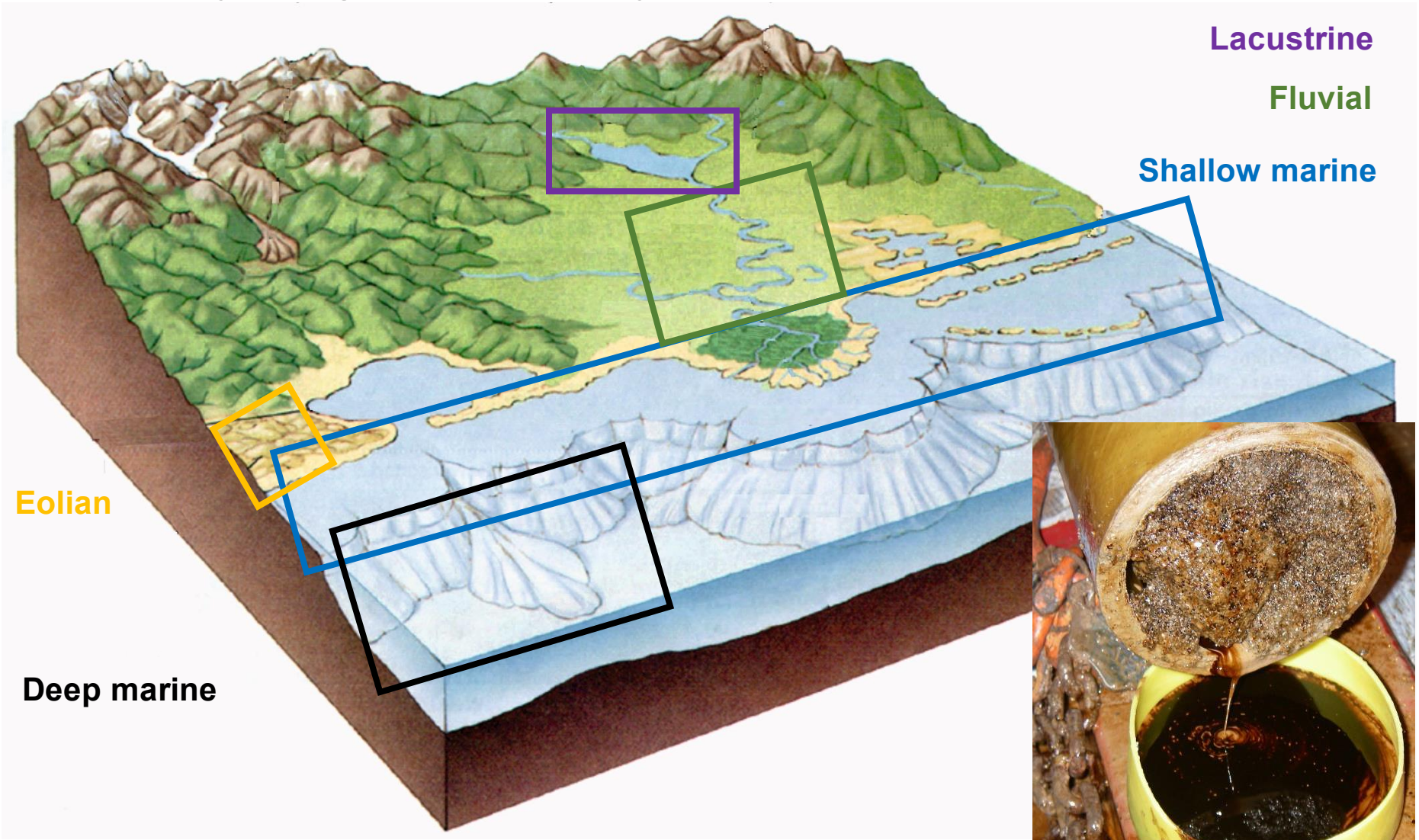


Jatiault 2017



Reservoir rock
Deposition

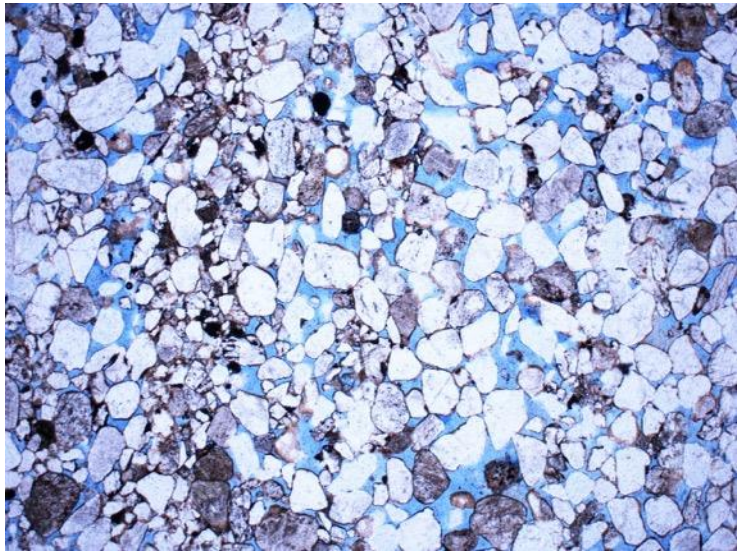
- Deposition and secondary modification of sedimentary rock: porosity and permeability (storage and flow capacity)



Reservoir rock
Deposition

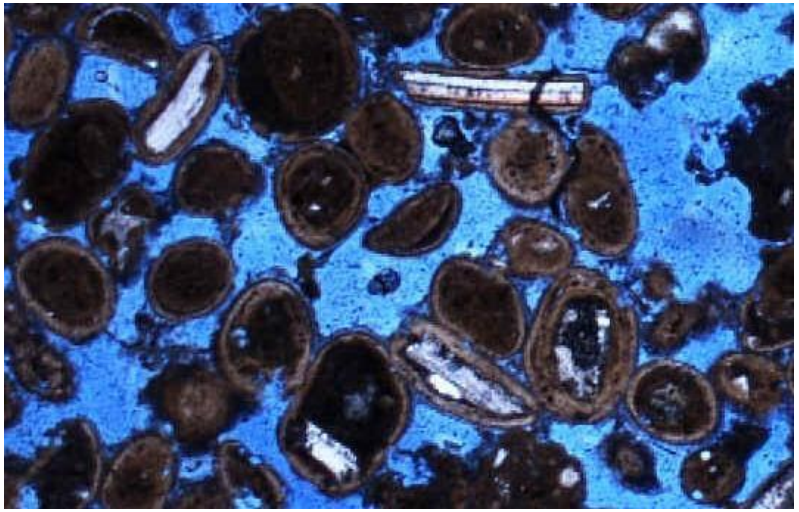
Sandstone





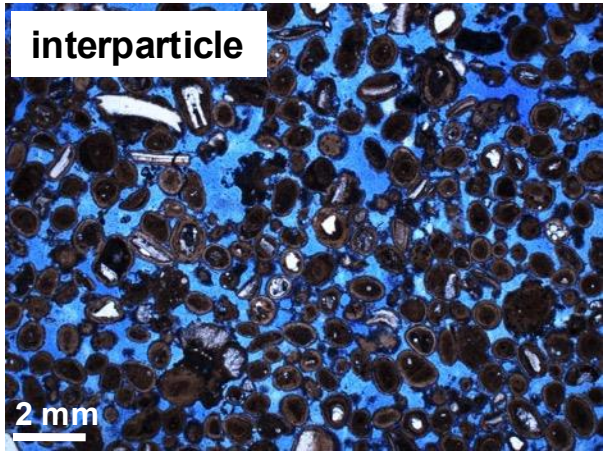
Limestone

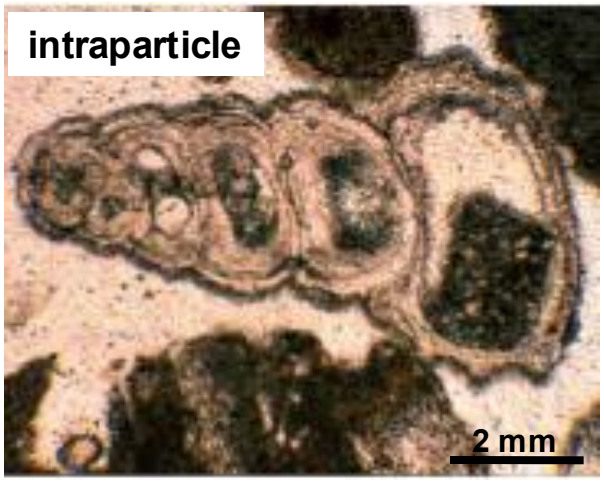




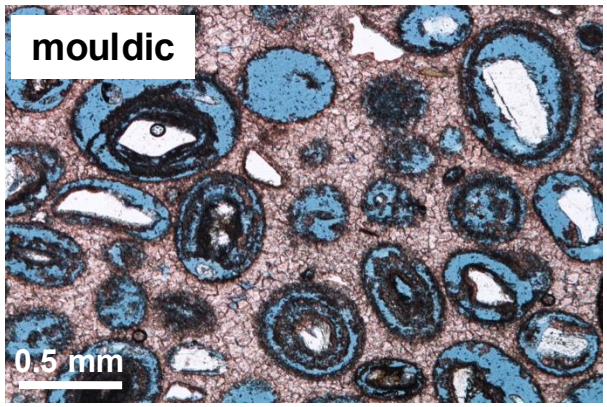
Reservoir rock
Pore network

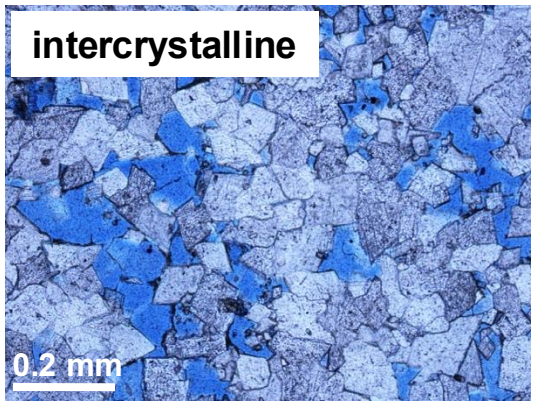
primary

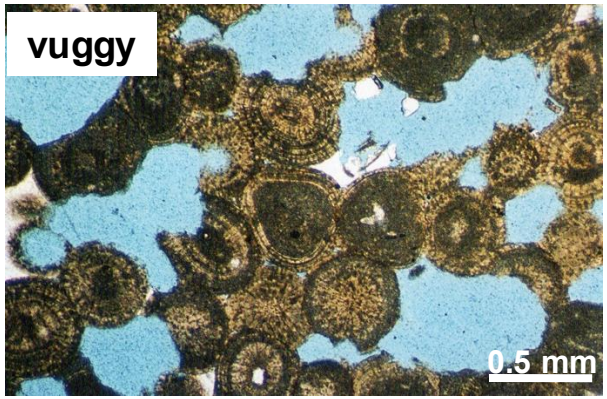


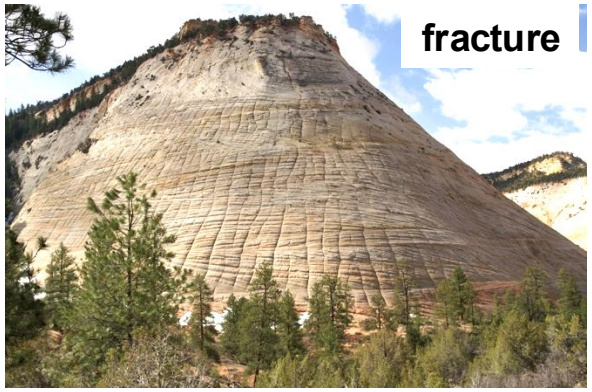


secondary



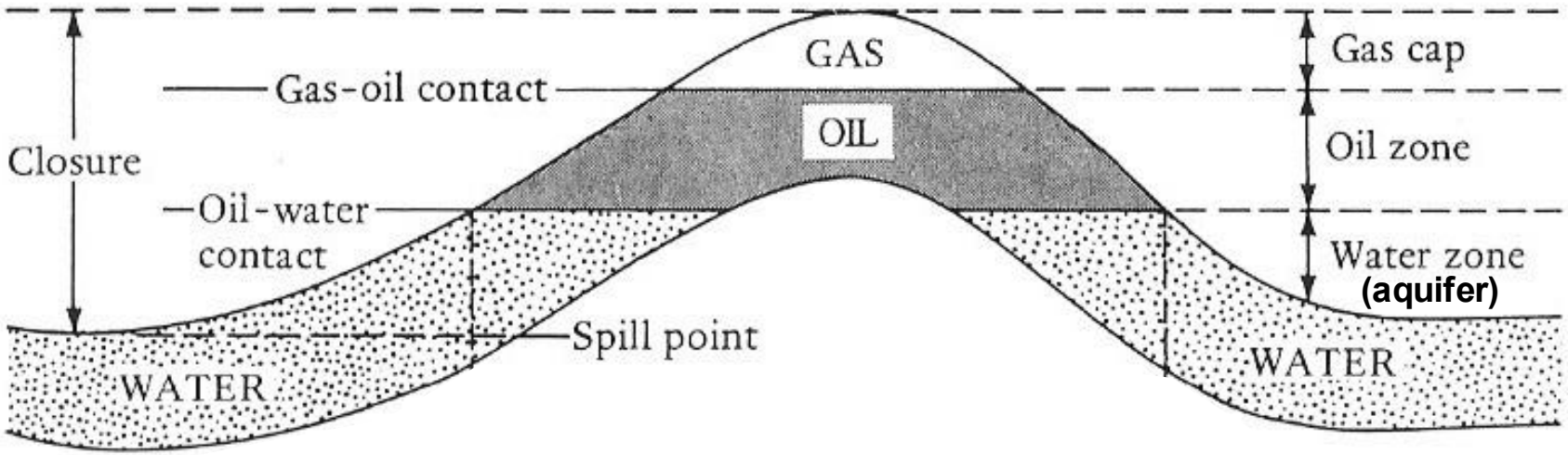






- Primary porosity reflects depositional character and controls on deposition (setting, energy level at deposition)
- Secondary porosity reflects diagenetic processes (cementation, compaction, dissolution….)

Trap
Nomenclature

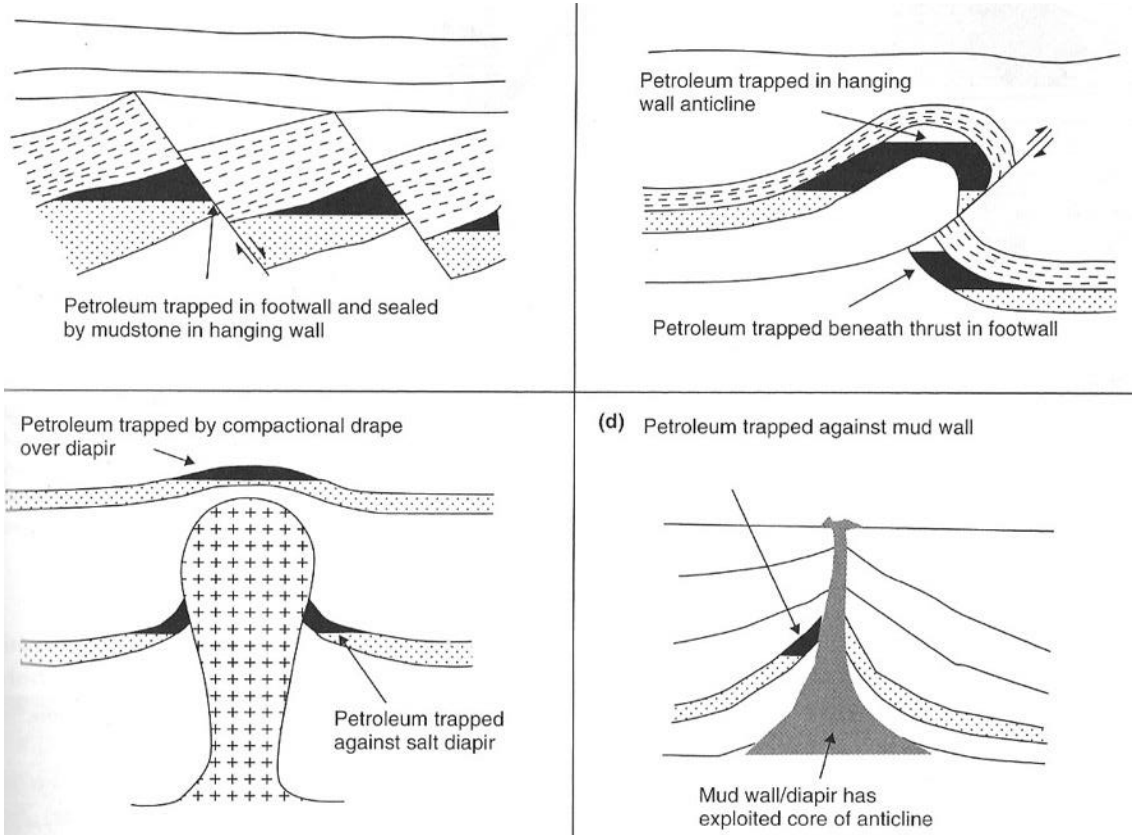


Selley 1998

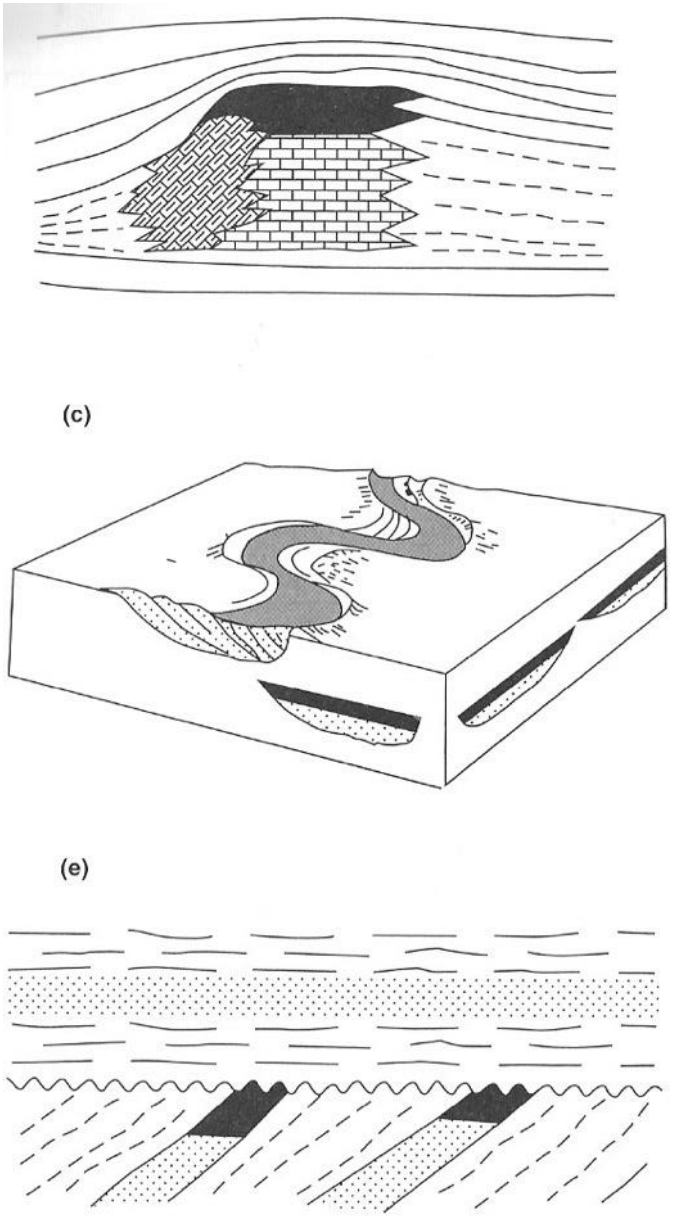
- **Traps provide a convex-up combination of reservoir and seal to keep hydrocarbons from moving up to surface**
- **Traps can be structural or stratigraphic**
- **Prediction requires structural geology, sedimentology and stratigraphy**

Trap
Classification

Structural traps

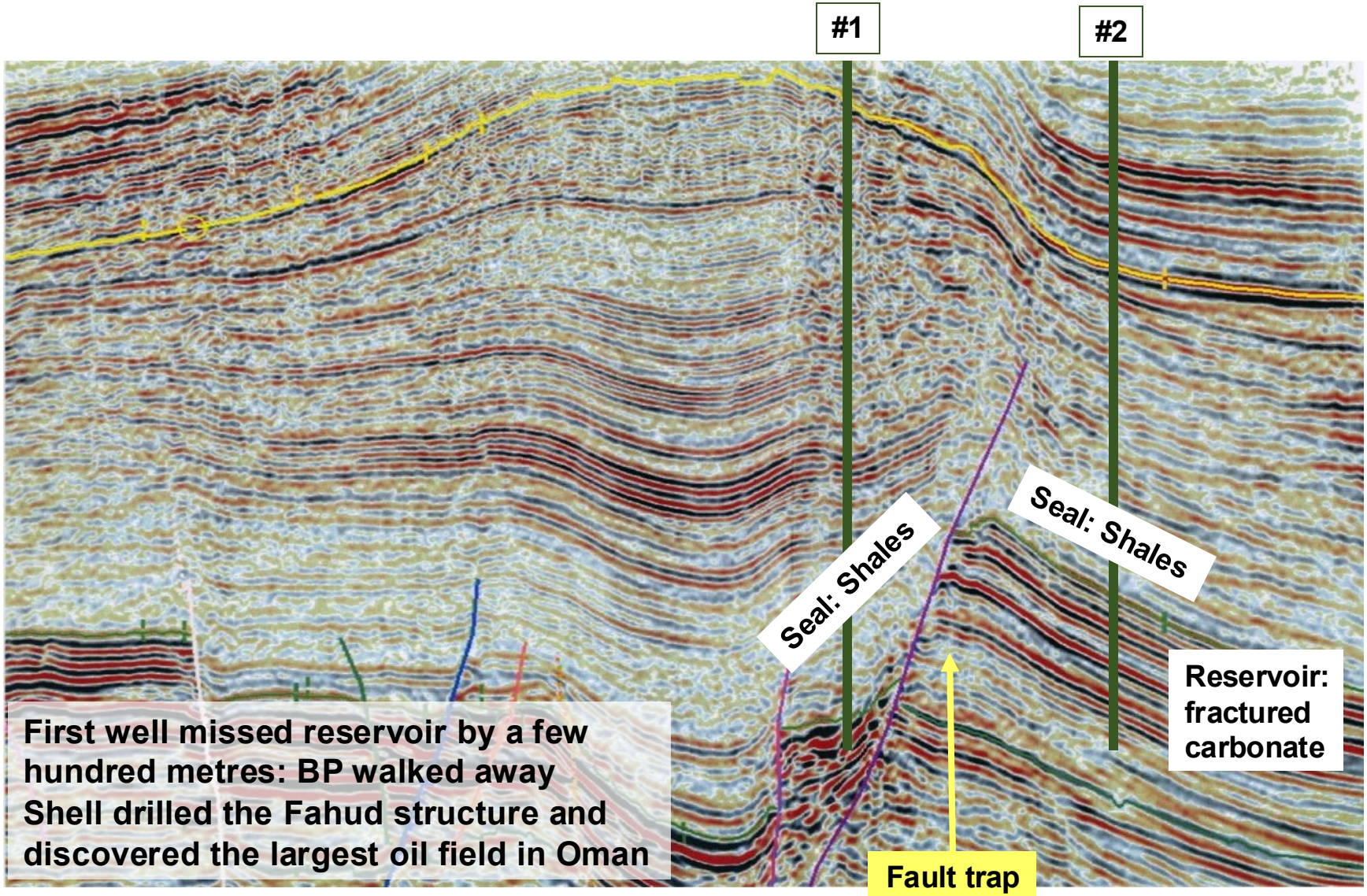


Stratigraphic traps

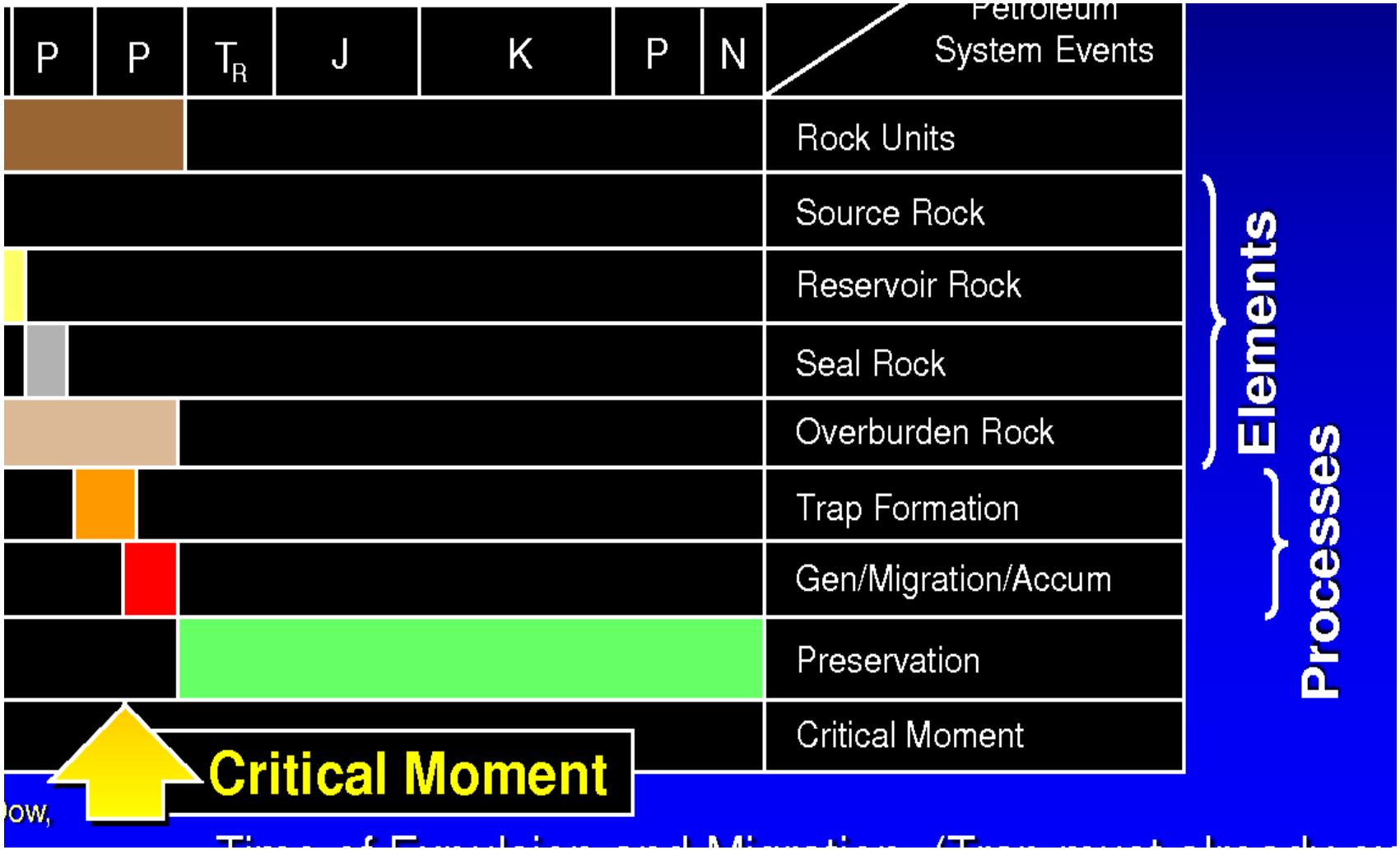


Gluyas 2011

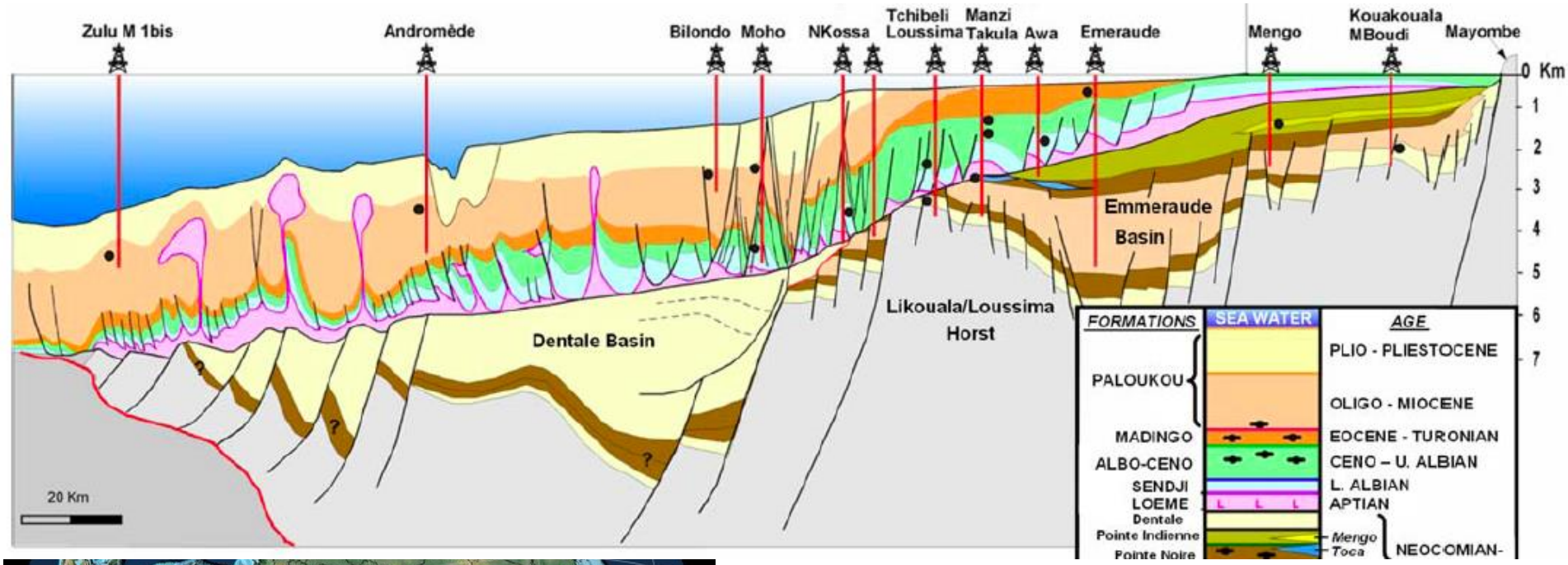
Trap
Example



Petroleum system



Petroleum system
West Africa case study

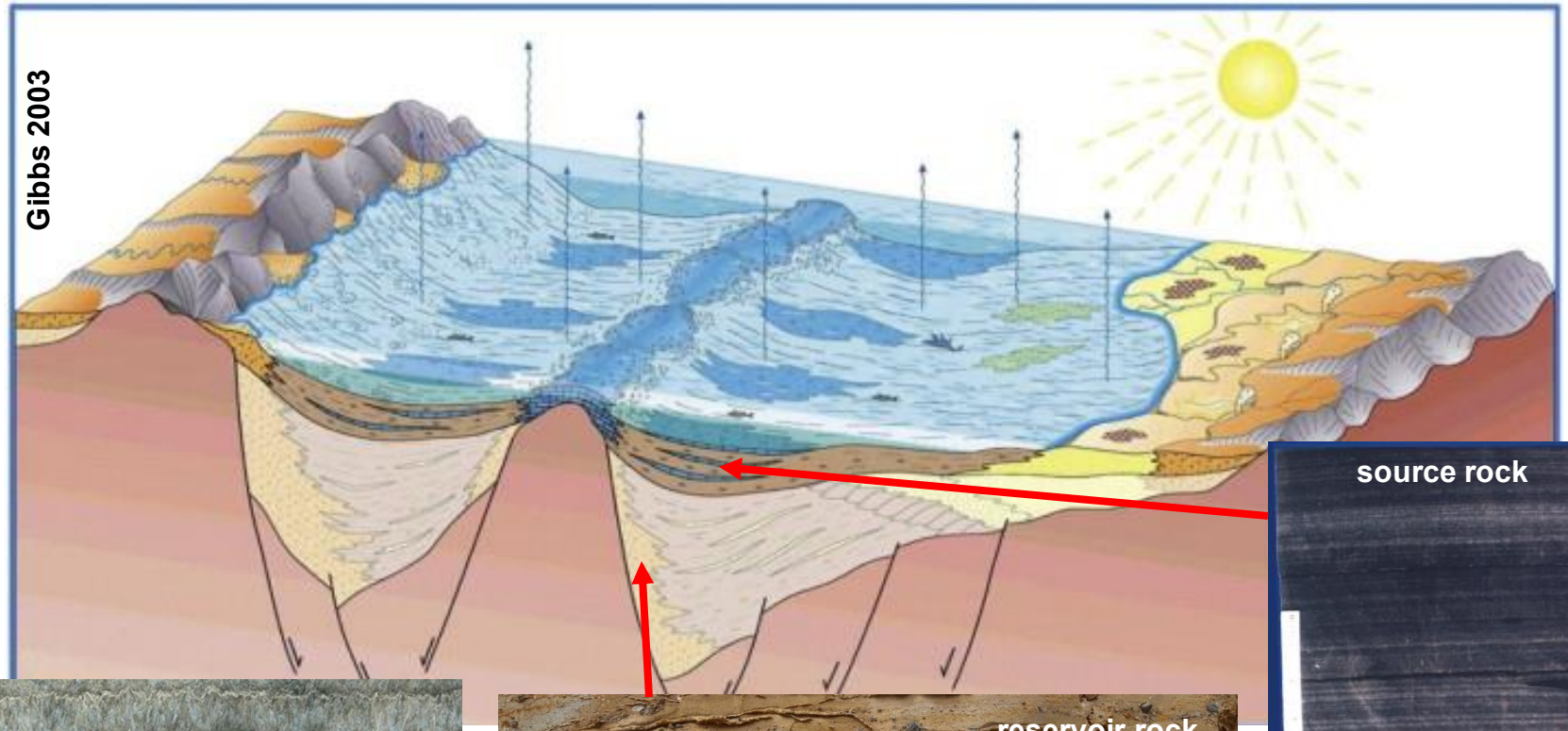


Wonham 2010



- **Lower Cretaceous syn-rift contains:**
  - **Lacustrine source rock**
  - **Continental sandstone reservoir rocks**
  - **Marine evaporite seals**
  - **Faulting provides traps**

Petroleum system
West Africa case study









Harris (2000)

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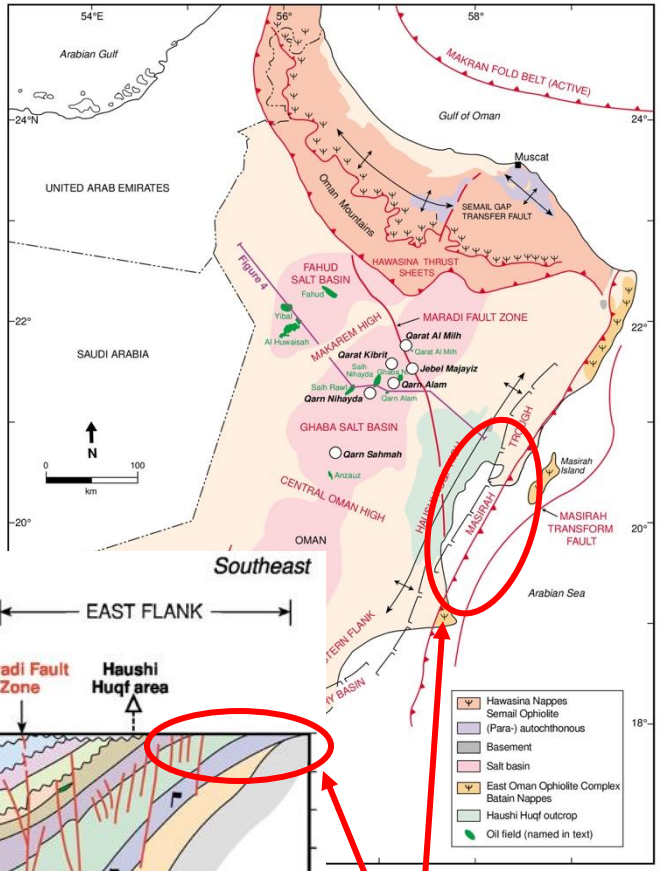
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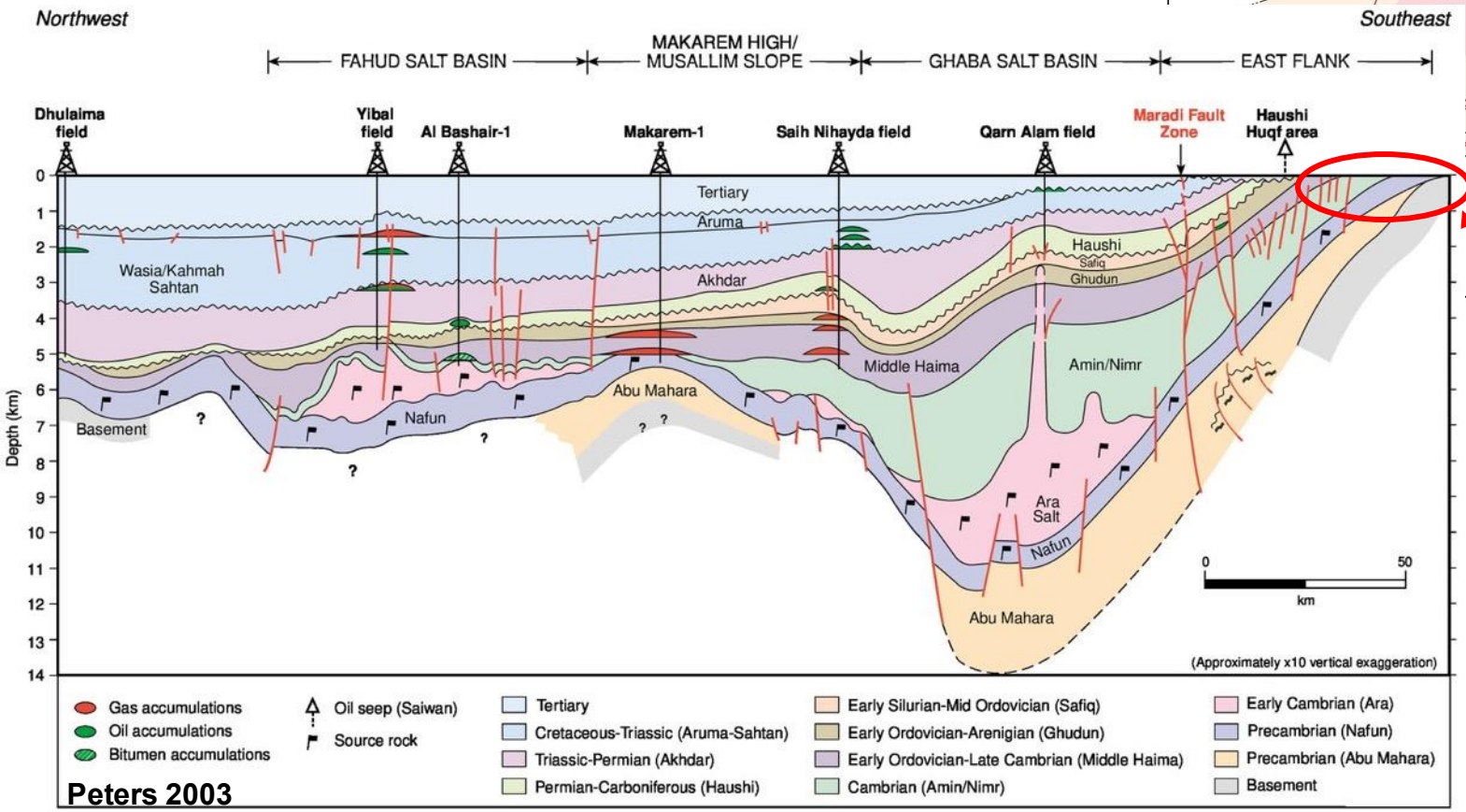
Lecture 8-3: Petroleum exploration and production



Exploration
Regional context

- Basin-margin outcrop studies to provide sedimentary and structural context
- Global/regional patterns of sedimentation (reservoirs, source rocks)



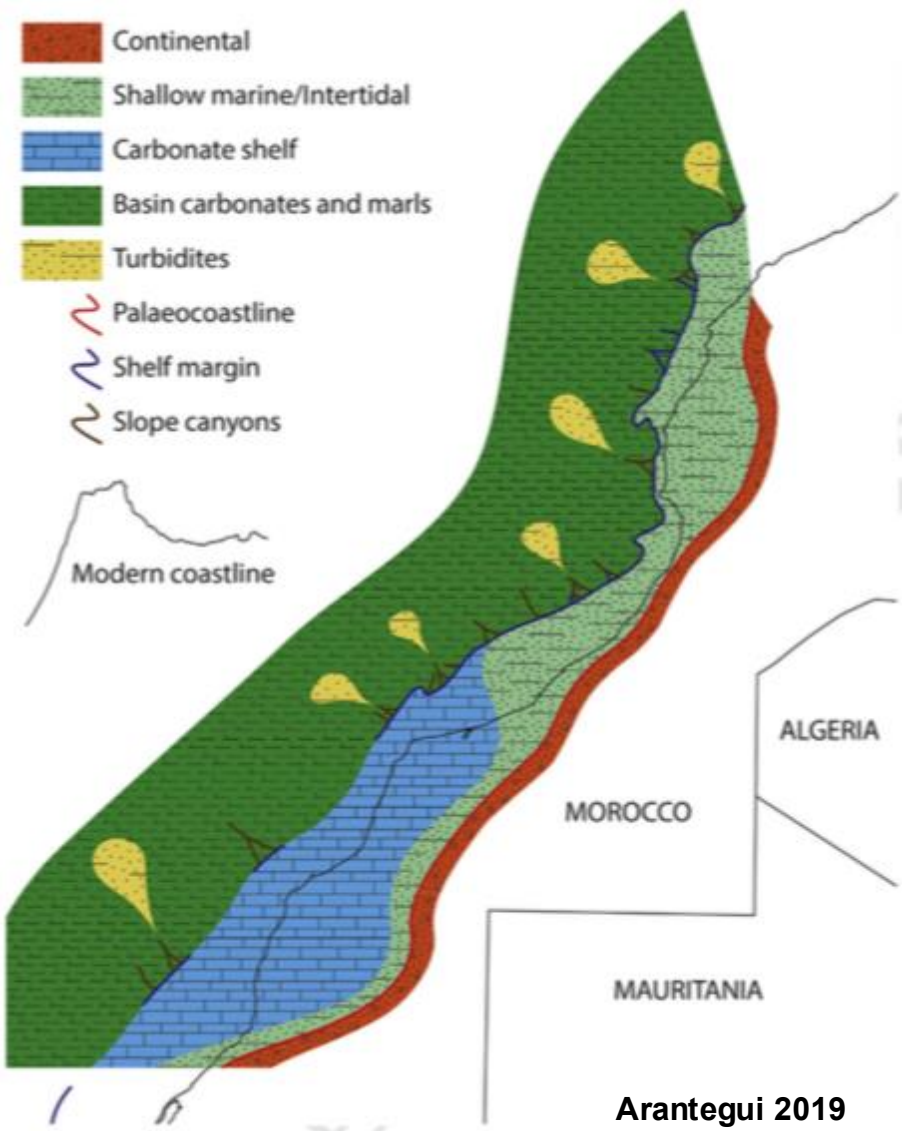


Peters 2003

outcrops

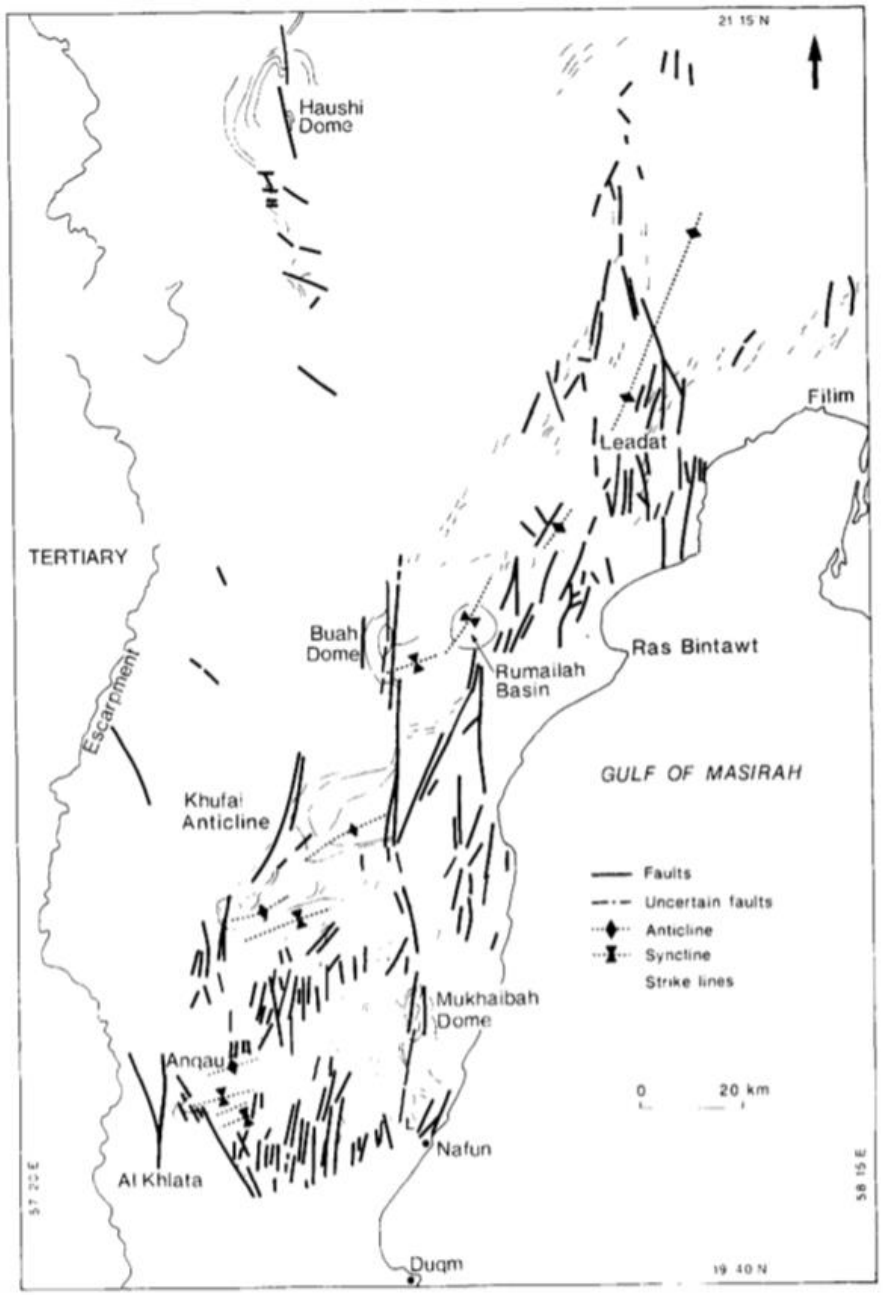
Exploration
Regional context

Analysis of sedimentation patterns



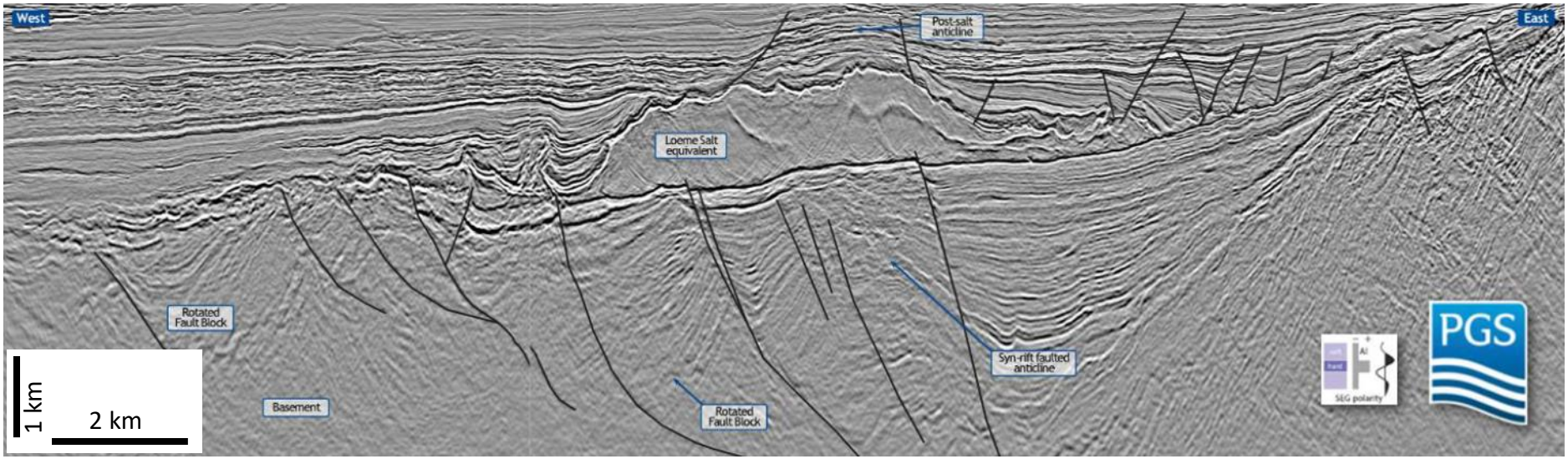
Arantegui 2019

Structural analysis



Ries 1990

Exploration
Geophysics

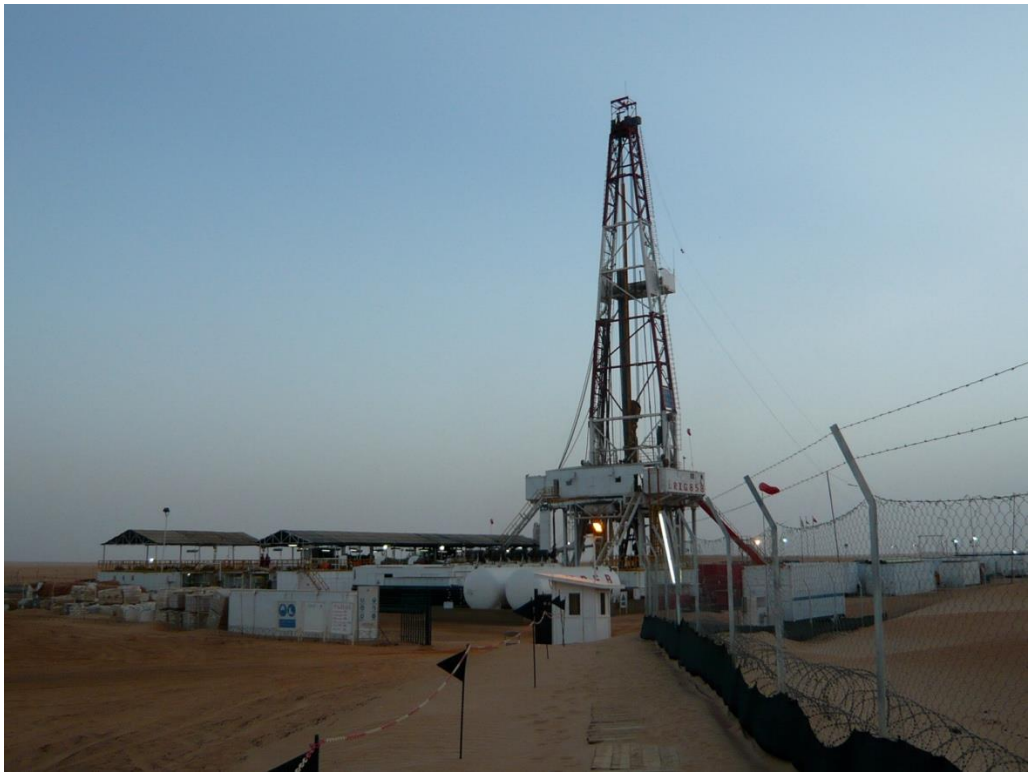


Reid 2016

- Seismic: large-scale structure of subsurface, sedimentary fill history of basin, identification of traps
- Magnetics and density: structural grain in subsurface
- Airborne detection of surface seeps

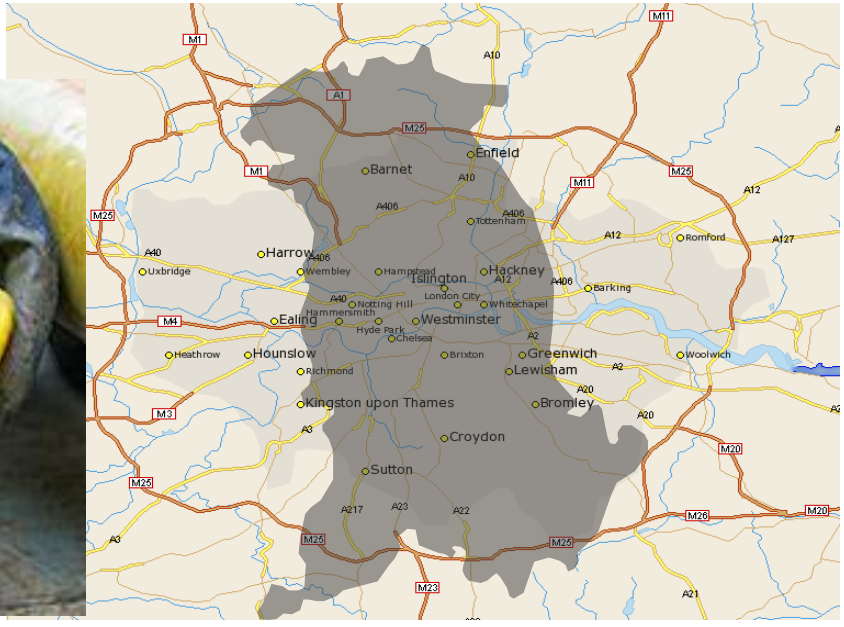
Exploration
Drilling

- Wells provide an essentially 1D view of subsurface
- Wells can be vertical and horizontal ('geosteering')
- Routine borehole lengths <8 km









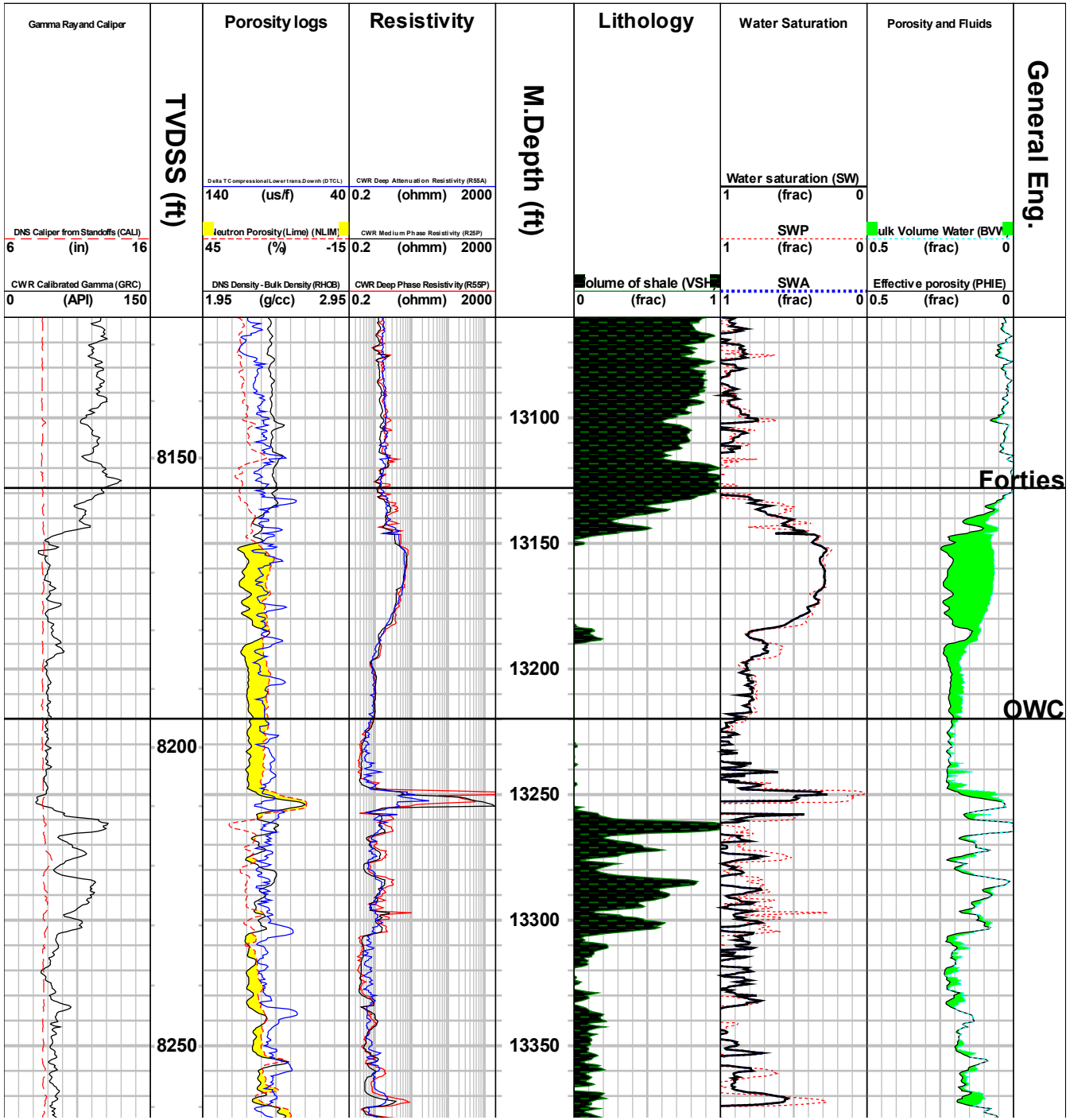
Source: Petroceltic

Exploration
Drilling



Source: www.expro.com

- Data collection during drilling via drill cuttings
- Wireline logging essential to obtain physical parameters describing subsurface around well



Production
Oil/gas fields





- Production wells, pipelines
- Minimizing environmental foot print
  - EIA
  - Remote subsea operations
  - Horizontal drilling
- Ongoing geological reservoir studies

- Conventional petroleum reserves: primary oil/gas recovery via free flow or secondary injection
- Unconventional petroleum reserves: oil/gas shales ('fracking'), tar sands



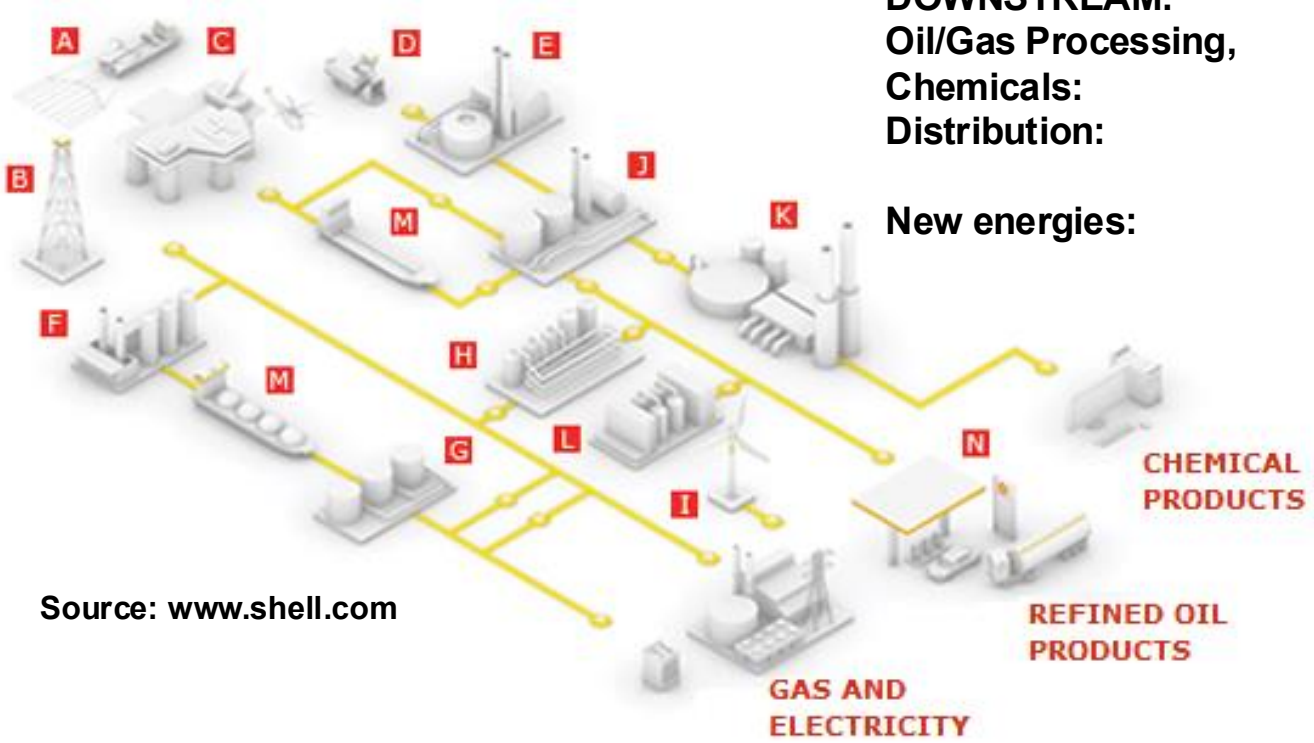
Source: www.energy4me.org

Production
Oil company structure

UPSTREAM
Exploration, Appraisal: finding oil and gas
Production: producing oil and gas

DOWNSTREAM:
Oil/Gas Processing, Chemicals: Refining & petrochemical manufacture
Distribution: Selling oil and gas products

New energies: wind, solar, geothermal, CCS



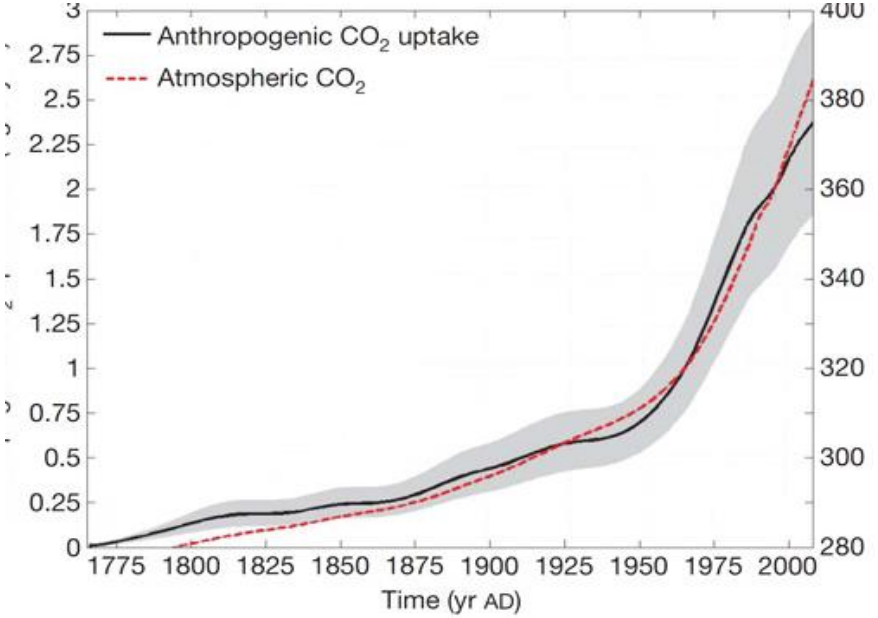
Source: www.shell.com

UPSTREAM	DOWNSTREAM
A - Exploring for oil and gas B - Developing fields C - Producing oil and gas D - Mining oil sands E - Extracting bitumen F - Liquefying gas by cooling (LNG) G - Regasifying LNG H - Converting gas to liquid products (GTL) I - Generating wind energy	J - Refining oil into fuels and lubricants K - Producing petrochemicals L - Developing biofuels M - Trading N - Retail sales * Managing CO2 emissions * Supply and distribution * Business-to-business sales

Environmental aspects

- All industrial activity has an environmental impact – generally negative (e.g. poisons are produced, damage is done, noise and harm to life, if not managed properly).
- The *potential* harm by industry activity is enormous – cf 2010 incident in Gulf of Mexico.
- The oil industry is intensely regulated for HSE (health, safety and environment). However, bad HSE practice is bad for business. Therefore, the oil industry often leads governments in safety and environmental practice.
- Biggest aspect to environmental impact – global warming and climate change: by producing and using petroleum we create an imbalance in the natural carbon cycle.
- Fossils fuels are unlikely to disappear quickly due to our reliance on them.
- Fossil fuels are still relatively cheap, and a means for many developing countries to reduce poverty - but corruption is also often associated with petroleum production and export.





Source: http://www.co2science.org

Petroleum deposits
Key learning points week 8

- Petroleum (hydrocarbons) form the basis of our current economy as they provide a relatively cheap and versatile fuel source. Related greenhouse gas emissions do however pose serious threats to the environment and climate.
- Deposits can be viewed in the framework of the petroleum system, which requires:
  - Matured source rock (influenced by productivity, preservation, dilution, composition, temperature and pressure)
  - Migration of hydrocarbon fluids
  - Accumulation of fluids in a trap (consists of porous/permeable reservoir rock and impermeable seal)
- Detailed structural, (organic) geochemical and sedimentary studies are required to identify all components of a working petroleum system
- Exploration passes through detailed regional knowledge, seismic studies, and drilling